|INCH-POUND|

MIL-M-28787/511
23 JUNE 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

IOC COMMON BUS CONTROL MODULE, KEY CODE BWF

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Input/Output Controller (IOC) common bus control module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:26Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.257 failures per million operating hours for class I or 2.703 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be BWF.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:26Z
Check the source to verify that this is the currentversion before use.

TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	--------Module shall pass all remaining tests----------						
D1-D8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D9	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D10	Ibd	V	S	0.0	0.6	0.0	0.6	0.0	0.6	mA
D11	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D12-D14	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D15-D18	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D19	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D20,D21	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D22	Ibd	V	S	-0.12	0.32	-0.12	0.32	-0.12	0.32	mA
D23	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D24-D27	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D28	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D29	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D30	Ibd	V	S	0.0	0.7	0.0	0.7	0.0	0.7	mA
D31	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D32	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D33	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D34	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D35	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D36	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D37	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D38	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D39	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D40	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D41	Ibd	V	S	0.0	7.1	0.0	7.1	0.0	7.1	mA

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ∓ 5°C		Class II 4/ -55 ± 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
E1-E5	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E6	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E7-E11	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E12,E13	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E14	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E15-E31	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E32	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E33	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E34	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E35	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E36	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E37,E38	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E39	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E40	Ios	VI	S	-230.0	-60.0	-230.0	-60.0	-230.0	-60.0	mA
E41-E43	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E44-E49	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	mA
F1-F8	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F9	Iih	VII	X	-1.2	0.04	-1 2	0.04	-1.2	0.04	mA
F10	Iih	VII	X	-3.6	0.12	-3.6	0.12	-3.6	0.12	mA
F11	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F12	Iih	VII	X	-2.2	0.04	-2.2	0.04	-2.2	0.04	mA
F13,F14	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F15,F16	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F17,F18	Iih	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F19	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F20,F21	Iih	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F22,F23	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F24	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F25	Iih	VII	X	-1.6	0.02	-1.6	0.02	-1.6	0.02	mA
F26-F28	Iih	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F29	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F30,F31	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F32	IIh	VII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
F33	IIh	VII	X	-6.63	-2.83	-6.63	-2.83	-6.63	-2.83	mA
F34	IIh	VII	X	-6.83	-2.85	-6.83	-2.85	-6.83	-2.85	mA
F35-F38	IIh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F39	IIh	VII	X	-2.8	0.05	-2.8	0.05	-2.8	0.05	mA
F40	IIh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F41	IIh	VII	X	-4.4	0.18	-4.4	0.18	-4.4	0.18	mA
F42	IIh	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
F43	IIh	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F44	IIh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F45	IIh	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F46	IIh	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F47	IIh	VII	X	-3.8	0.06	-3.8	0.06	-3.8	0.06	mA
F48	IIh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F49	IIh	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F50	IIh	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F51	IIh	VII	X	-0.8	0.1	-0.8	0.1	-0.8	0.1	mA
F52	IIh	VII	X	-6.6	0.37	-6.6	0.37	-6.6	0.37	mA
G1-G8	III	VIII	X	-0.8	0.02	-0.6	0.02	-0.6	0.02	mA
G9	III	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G10	III	VIII	X	-3.6	0.12	-3.6	0.12	-3.6	0.12	mA
G11	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G12	III	VIII	X	-2.2	0.04	-2.2	0.04	-2.2	0.04	mA
G13,G14	III	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G15,G16	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G17,G18	III	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G19	III	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G20,G21	III	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G22,G23	III	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G24	III	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G25	III	VIII	X	-1.6	0.02	-1.6	0.02	-1.6	0.02	mA
G26-G28	III	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G29	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G30,G31	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G32	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G33	Iil	VIII	X	-6.63	-2.83	-6.63	-2.83	-6.63	-2.83	mA
G34	Iil	VIII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
G35-G38	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G39	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G40	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G41	Iil	VIII	X	-4.4	0.18	-4.4	0.18	-4.4	0.18	mA
G42	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G43	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G44	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G45	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G46	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G47	Iil	VIII	X	-3.0	0.06	-3.0	0.06	-3.0	0.06	mA
G48	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G49	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G50	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G51	Iil	VIII	X	-0.8	0.1	-0.8	0.1	-0.8	0.1	mA
G52	Iil	VIII	X	-6.6	0.37	-6.6	0.37	-6.6	0.37	mA
H1-H49	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J48	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K6	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K7-K12	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1-L39	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	6.3	11.7	6.3	11.7	6.3	11.7	uF
M2	PSFC	XIII	S	5.6	10.4	5.6	10.4	5.6	10.4	uF

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
N	Icc	XIV	X	535	1284	535	1284	535	1284	mA
P1,P2	Functional	XV	X	----------See table XV----------						
Q1,Q2	Dynamic	XVI	S	----------See table XVI----------						
Q3,Q4	Dynamic	XVI	Q	----------See table XVI----------						
R1	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R2-R4	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R5	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R6	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R7,R8	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R9	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R10	C1/o	XVIII	Q	10	175	10	175	10	175	pF
R11	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R12	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R13	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R14	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R15-R17	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R18-R21	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R22	C1/o	XVIII	Q	6	100	6	100	6	100	pF
R23	C1/o	XVIII	Q	2	100	2	100	2	100	pF
R24,R25	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R26	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R27	C1/o	XVIII	Q	2	100	2	100	2	100	pF
R28	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R29	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R30	C1/o	XVIII	Q	14	175	14	175	14	175	pF
R31	C1/o	XVIII	Q	4	50	4	50	4	50	pF
R32-R34	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R35	C1/o	XVIII	Q	2	100	2	100	2	100	pF
R36	C1/o	XVIII	Q	6	125	6	125	6	125	pF
R37	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R38	C1/o	XVIII	Q	4	125	4	125	4	125	pF
R39	C1/o	XVIII	Q	2	50	2	50	2	50	pF

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R40	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R41	C1/o	XVIII	Q	16	150	16	150	16	150	pF
R42	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R43	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R44-R48	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R49	C1/o	XVIII	Q	2	100	2	100	2	100	pF
R50-R52	C1/o	XVIII	Q	2	75	2	75	2	75	pF
S1,S2	Input clamp diode latch up test	XIX	Q	-1.2	2.4	-1.2	2.4	-1.2	2.4	V
T1,T2	PSTA	XX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T18:26Z
Check the source to verify that this is the current version before use.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/511-1 for class I and MIL-M-28787/511-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

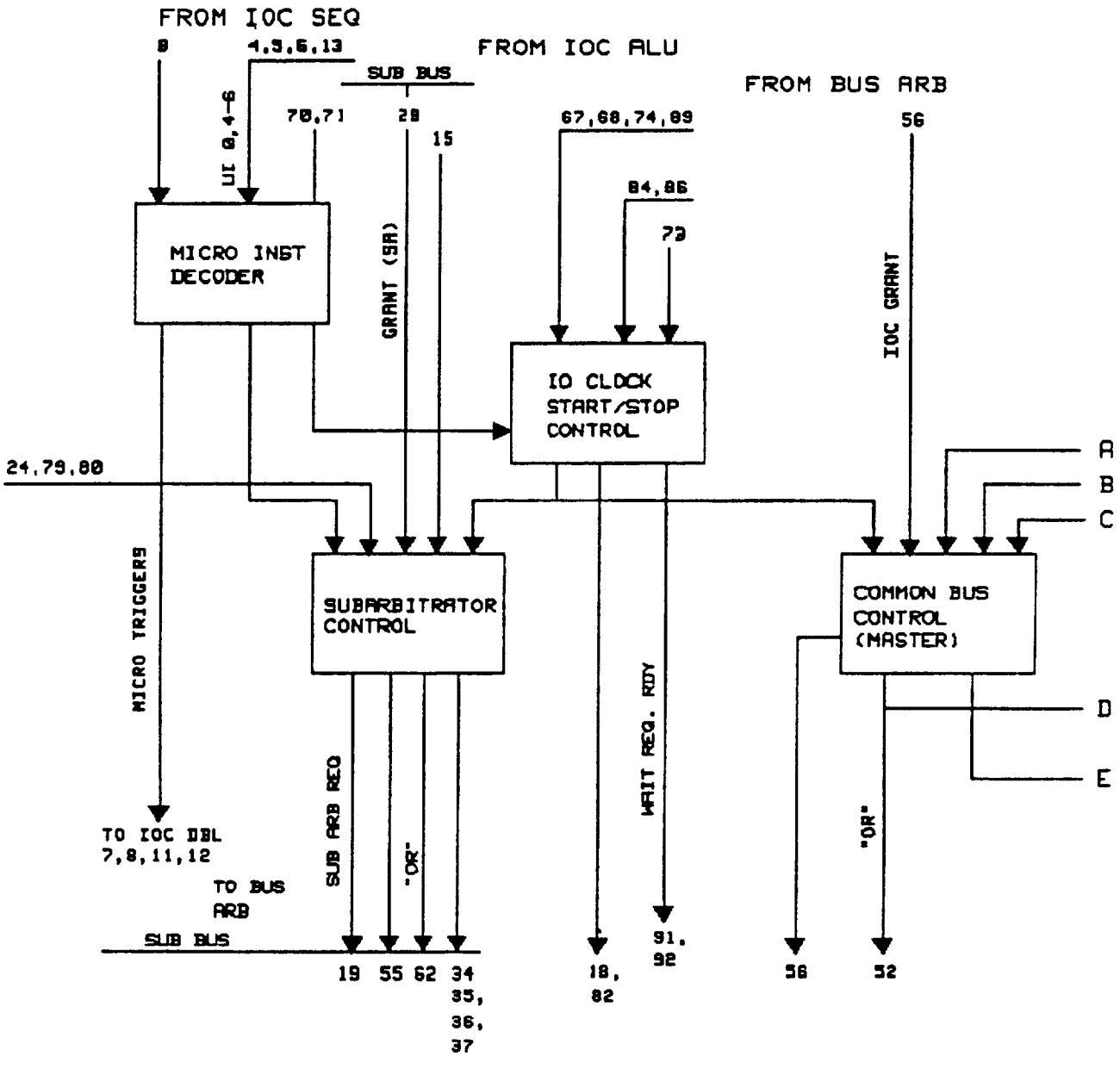


FIGURE 1. BWF functional block diagram.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

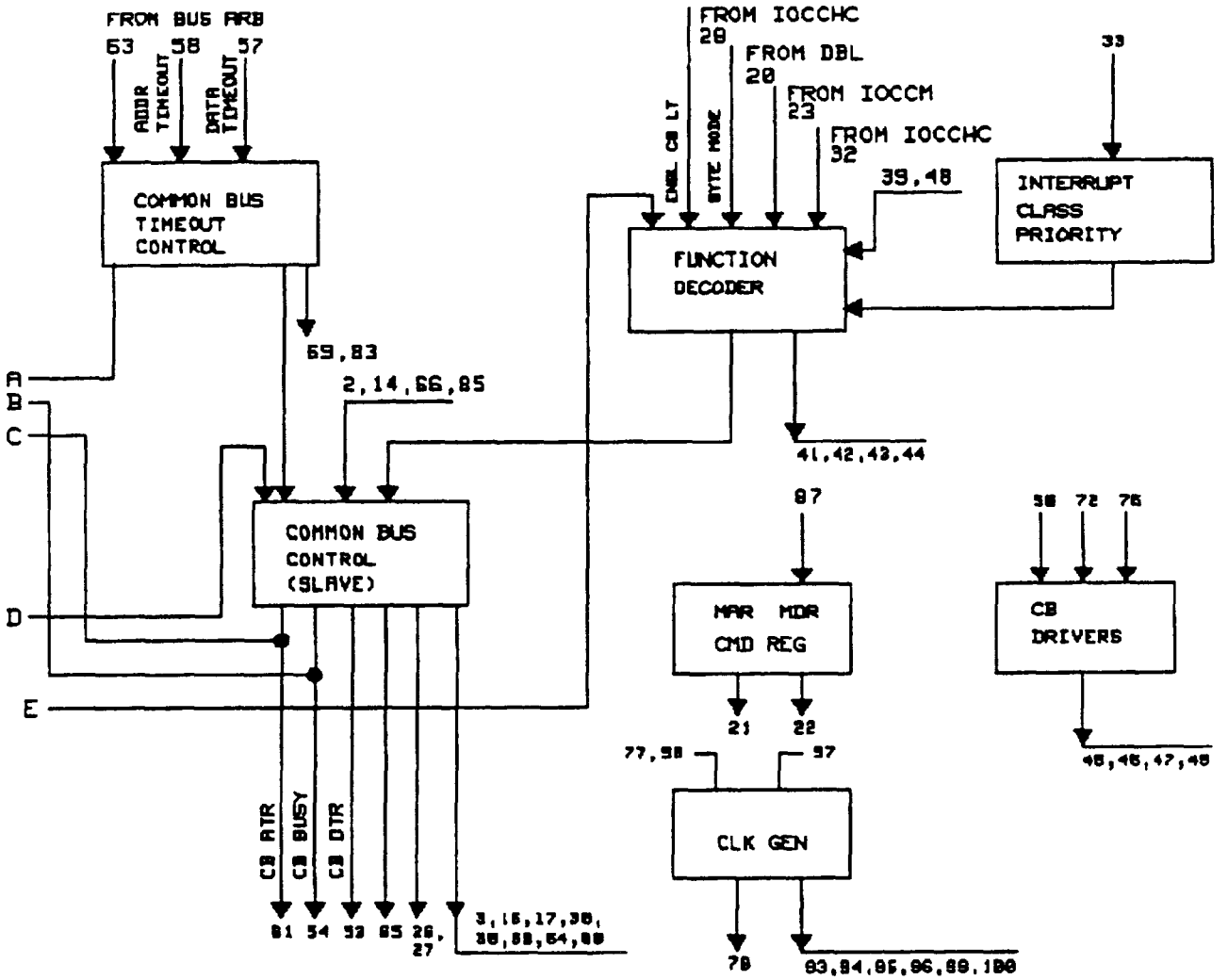


FIGURE 1. BWF functional block diagram - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

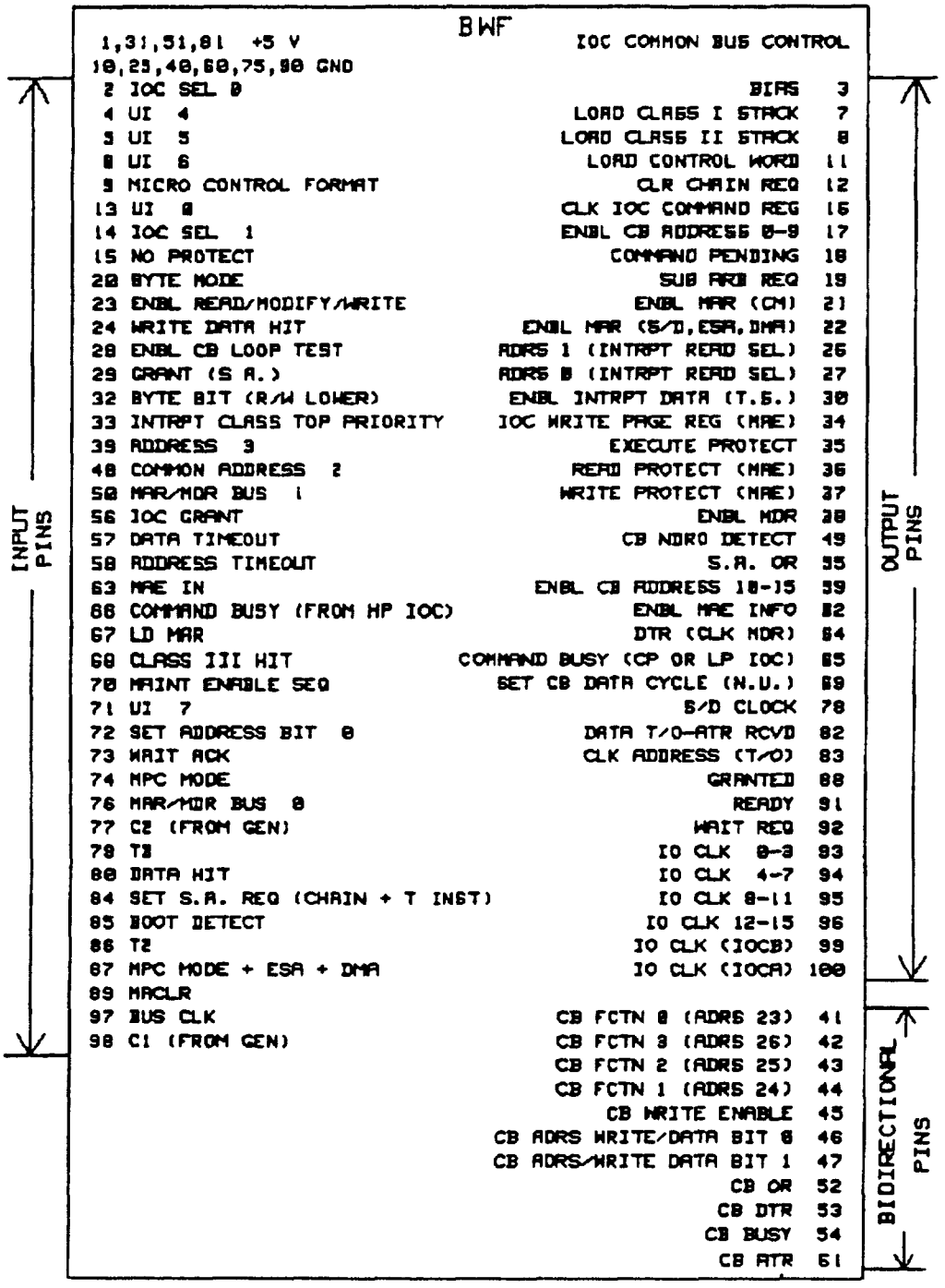


FIGURE 2. BWF pin assignment.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:26Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L and S are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, L or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L and S where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature

Source: https://assist.dla.mil -- Downloaded: 20260410-20T18:26Z
Check the source to verify that this is the current version before use.

stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

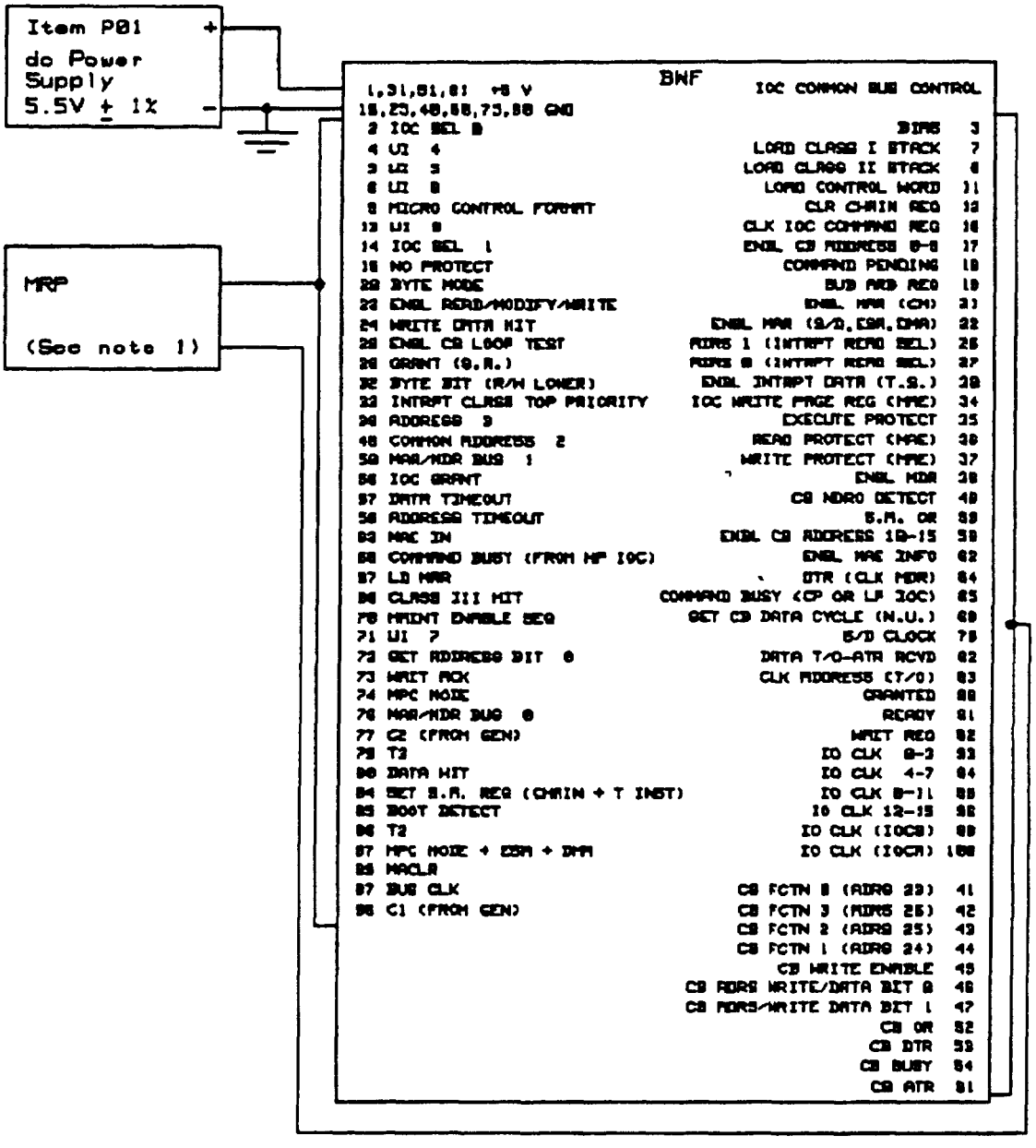
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.



NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. BWF operating temperature and component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

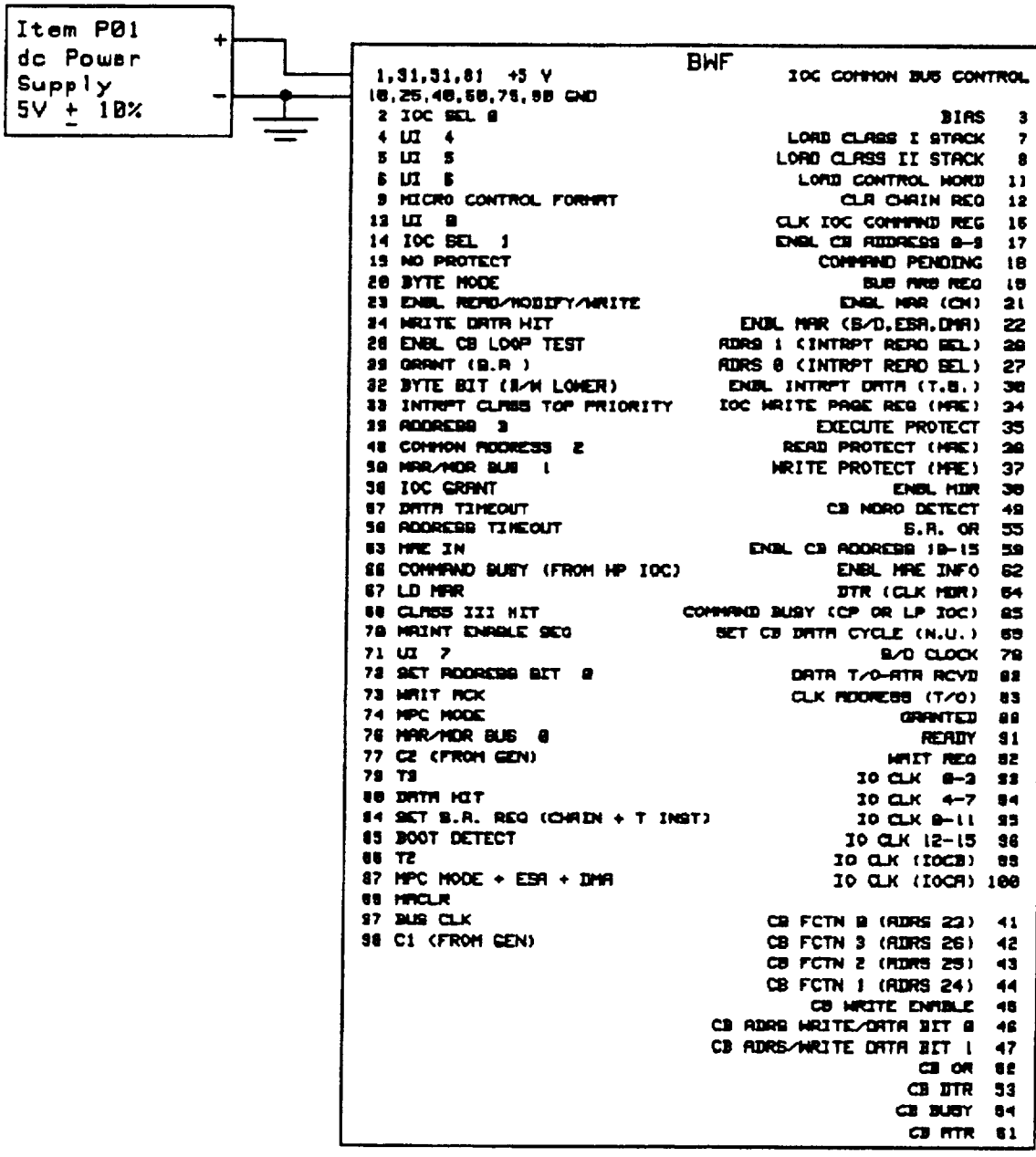
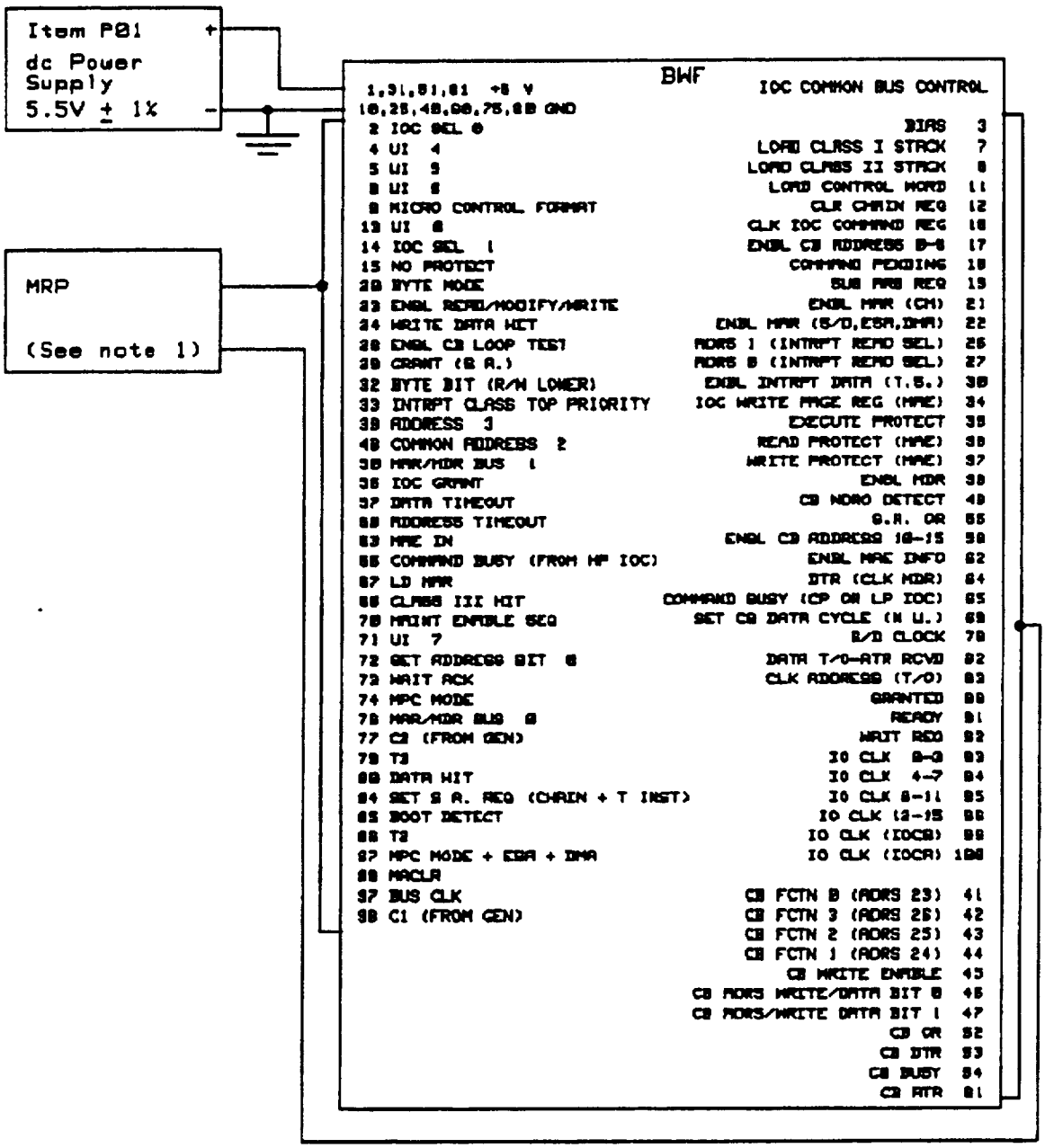


FIGURE 4. BWF vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.



NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. BWF life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 7.062 W (max); 4.735 W (typical).

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IOC SEL 0	-1.2	0.4	2.4	-0.6	0.02	0.1	75
UI 4	-1.2	0.4	2.4	-0.6	0.02	0.1	50
UI 5	-1.2	0.4	2.4	-0.6	0.02	0.1	50
UI 6	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MCO CTRL FMT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
UI 0	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IOC SEL 1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
NO PROTECT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
BYTE MODE	-1.2	0.4	2.4	-1.2	0.04	0.2	75
ENBL R/M/W	-1.2	0.4	2.4	-3.6	0.12	0.6	175
WR DATA HIT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
ENBL CB LT	-1.2	0.4	2.4	-2.2	0.04	0.2	125
GRANT (S.A.)	-1.2	0.4	2.4	-1.2	0.04	0.2	75
BTE/BT (R/W)	-1.2	0.4	2.4	-1.2	0.04	0.2	100

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
INT C TP PRI	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ADDRESS 3	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COMM ADDR 2	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MAR/MDR B 1	-1.2	0.4	2.4	-1.6	0.02	0.1	50
IOC GRANT	-1.2	0.4	2.4	-2.0	0.05	1.0	50
DATA TIMEOUT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
ADDR TIMEOUT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MAE IN	NA	0.4	2.4	-6.63	-2.83	0.32	100
COMMAND BUSY	NA	0.4	2.4	-6.03	-2.85	0.22	100
LD MAR	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CLS III HIT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MAIN ENBL SQ	-1.2	0.4	2.4	-0.6	0.02	0.1	75
UI 7	-1.2	0.4	2.4	-0.6	0.02	0.1	100
ST ADDR BT 0	-1.2	0.4	2.4	-2.0	0.05	1.0	50

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
WAIT ACK	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MPC MODE	-1.2	0.4	2.4	-4.4	0.18	0.7	175
MAR/MDR B 0	-1.2	0.4	2.4	-4.0	0.1	2.0	50
C2 (FRM GEN)	-1.2	0.4	2.4	-2.0	0.05	1.0	50
T3	-1.2	0.4	2.4	-0.6	0.02	0.1	50
DATA HIT	-1.2	0.4	2.4	-2.0	0.05	1.0	50
SET S.A. REQ	-1.2	0.4	2.4	-0.8	0.04	0.2	100
BOOT DETECT	-1.2	0.4	2.4	-3.8	0.06	0.3	125
T2	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MPC M+E+D	-1.2	0.4	2.4	-1.2	0.04	0.2	125
MACLR	-1.2	0.4	2.4	-0.6	0.02	0.1	50
BUS CLK	-1.2	0.4	2.4	-0.8	0.1	2.0	75
C1 (FRM GEN)	-1.2	0.4	2.4	-6.6	0.37	7.1	150
CB FCTN 0	NA	0.4	2.4	-0.65	0.07	NA	75

Source: https://assist.dla.mil -- Downloaded: 20261O-20T18:26Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CB FCTN 3	NA	0.4	2.4	-0.65	0.07	NA	100
CB FCTN 2	NA	0.4	2.4	-0.25	0.07	NA	75
CB FCTN 1	NA	0.4	2.4	-0.65	0.07	NA	75
CB WR ENBL	NA	0.4	2.4	-0.65	0.07	NA	75
CB AD/WR D0	NA	0.4	2.4	-0.25	0.07	NA	75
CB AD/WR D1	NA	0.4	2.4	-0.25	0.07	NA	75
CB OR	NA	0.4	2.4	-0.65	0.07	NA	100
CB DTR	NA	0.4	2.4	-0.65	0.07	NA	75
CB BUSY	NA	0.4	2.4	-0.65	0.07	NA	75
CB ATR	NA	0.4	2.4	-0.65	0.07	NA	75

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
BIAS	2.4	0.5	-0.4	NA	NA	NA	-60.0	NA
LD CL I STK	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
LD CL II STK	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
LD CTRL WORD	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CLR CHAIN RQ	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CK IOC CM RG	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
ENBL CB 0-9	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
COMM PEND	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SUB ARB REQ	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ENB MAR (CM)	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ENB MR (S/D)	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ADRS 1 (IRS)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
ADRS 0 (IRS)	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
ENB INT DTA	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
IOC WR PG RG	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
EXEC PROTECT	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
RD PRO (MAE)	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
WR PRO (MAE)	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
ENBL MDR	2.4	0.5	-0.4	6.0	NA	NA	-100.0	NA
CB NDRO DTCT	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
S.A. OR	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
ENL CB 10-15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ENL MAE INFO	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
DTR (CK MDR)	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
COM BSY	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SET CB D CYC	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
S/D CLOCK	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
D T/O-ATR RD	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

Source: https://assist.dla.mil -- Downloaded: 20261O-20T18:26Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CLK AD (T/O)	2.4	0.5	-0.16	1.6	NA	NA	-60.0	NA
GRANTED	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
RDY	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
WAIT REQ	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
IO CLK 0-3	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
IO CLK 4-7	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
IO CLK 8-11	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
IO CLK 12-15	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
IO CK (IOCB)	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
IO CK (IOCA)	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
CB FCTN 0	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
CB FCTN 3	2.4	0.5	-0.4	6.0	NA	NA	-100.0	100
CB FCTN 2	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
CB FCTN 1	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CB WR ENBL	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
CB AD/WR D0	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
CB AD/WR D1	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
CB OR	2.4	0.5	-0.4	6.0	NA	NA	-100.0	100
CB DTR	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
CB BUSY	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75
CB ATR	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	GRANT (S.A.) to BUS CLK	23	23	23	--	--	--	nS	12
2	th	BUS CLK to GRANT (S.A.)	6	6	6	--	--	--	nS	12
3	tsu	COMM ADDR 2 to BUS CLK	20	20	20	--	--	--	nS	12
3	tsu	ADDRESS 3 to BUS CLK	20	20	20	--	--	--	ns	12
4	th	BUS CLK to COMM ADDR 2	26	26	26	--	--	--	nS	12
4	th	BUS CLK to ADDRESS 3	26	26	26	--	--	--	nS	12
5	tsu	IOC SEL 0 to BUS CLK	35	35	35	--	--	--	nS	12
5	tsu	IOC SEL 1 to BUS CLK	35	35	35	--	--	--	nS	12
6	th	BUS CLK to IOC SEL 0	3	3	3	--	--	--	nS	12
6	th	BUS CLK to IOC SEL 1	3	3	3	--	--	--	nS	12
7	tsu	MPC MODE to BUS CLK	23	23	23	--	--	--	nS	12
8	tsu	ENBL R/M/W to C1 (FRM GEN)	39	39	39	--	--	--	nS	12
9	th	C1 (FRM GEN) to ENBL R/M/W	3	3	3	--	--	--	nS	12
10	tpd	BUS CLK to SUB ARB REQ	--	--	--	27	27	27	nS	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

6.3.3 Class I and II dynamic parameters. - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	BUS CLK to CK IOC CM RG	--	--	--	39	39	39	nS	12
11	tpd	BUS CLK to COM BSY	--	--	--	39	39	39	nS	12
12	tpd	BUS CLK to ENB MAR (CM)	--	--	--	41	41	41	nS	12
12	tpd	BUS CLK to ENB MR (S/D)	--	--	--	41	41	41	nS	12
12	tpd	BUS CLK to ENBL MDR	--	--	--	41	41	41	nS	12
13	tpd	BUS CLK to CB ATR	--	--	--	34	34	34	nS	12
14	tpd	BUS CLK to ENBL CB 0-9	--	--	--	39	39	39	nS	12
15	tpd	BUS CLK to ADRS 0 (IRS)	--	--	--	61	61	61	nS	12
15	tpd	BUS CLK to ADRS 1 (IRS)	--	--	--	61	61	61	nS	12
16	tpd	GRANT (S.A.) to ENL MAE INFO	--	--	--	25	25	25	nS	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

6.4 Module function. The IOC common bus control module contains a sub-request generator, the function decoder, common bus master circuits, and common bus slave circuits. This module, residing on the common bus, provides the IOC with the ability to request the use of the common bus (master) or to receive commands/data from another master (slave). The function decoder receives common bus bits from the emulator control 2 module to determine the function the IOC is to perform in the slave mode. The sub-request generator sends out requests on the sub bus (sub-request) to the bus arbitrator module to use the common bus. Once the bus arbitrator has determined priority and acknowledged the request (sub-grant), the sub-request generator responds with a ready (originator ready) signal and the bus arbitrator initiates a common bus request. The common bus control (master) is triggered by the grant (IOC grant) from the bus arbitrator module to use the common bus. Address information is placed onto the common bus and the addressed device will respond with an acknowledge Addressed Terminator Ready (ATR). Data is then placed onto the common bus and the addressed device will respond with an acknowledge Data Terminator Ready (DTR) once the data has been accepted. The IOC becomes a slave to an operation when the Central Processor (CP) addresses the IOC with a master write function code. The IOC common bus control module will respond with an acknowledge from the common bus (slave) circuits. The address will be accepted and data will be placed onto the common bus by the master device. The IOC will then respond with an acknowledge and the cycle will end.

6.4.1 Microinstruction trigger decoder. The microinstruction trigger decoder receives microinstruction bits 6-4 to provide control signals to the IOC data bus lower modules and read, write, and stop signals to the clock start/stop circuits and the sub arbitrator. Enables are provided by microinstruction bit 0 and the trigger enable signal from the timeout control circuits.

6.4.2 Sub arbitrator control circuits. The sub arbitrator control circuits consist of a sub-request latch, a sub arbitrator originator ready latch, and a sub-grant detector. The IOC sub arbitrator control circuits send out a request (sub-request) to the bus arbitrator module for use of the common bus. Once the bus arbitrator module determines priority (IOC 0 highest, IOC 3 lowest), it sends out a sub-grant (Grant Source A (SA)) to the IOC allowing it to use Memory Address Expansion (MAE) (if installed). Once the SA Grant has been detected by the sub arbitrator control circuits, the SA originator ready signal is generated and transmitted to the bus arbitrator module.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:26Z
Check the source to verify that this is the current version before use.

Upon receipt of this signal, the bus arbitrator module initiates the request to use the common bus. The sub arbitrator request latch is reset by four signals applied to the "J" input of the flip-flop. These are:

(a) T Read.

(b) Set SA Req (data).

(c) Set SA Req Chain + T Inst.

(d) T Write and Not En Read/Mod/Write.

The sub arbitrator request latch output (sub-request) is sent to the bus arbitrator module (the signal is held low for 5 bus clock cycles or 250 ns). Once the bus arbitrator module has determined priority, it responds to the selected IOC with a sub arbitrator grant. The sub arbitrator grant is held low for two bus clock cycles (100 ns). After receipt, the sub arbitrator grant is detected and applied as an enable to the sub arbitrator control originator ready latch. The output of this latch is the Originator Ready (OR) signal which is transmitted back to the bus arbitration module. At the trailing edge of the OR signal, the bus arbitrator requests use of the common bus.

6.4.3 Common bus control (master). Upon receipt of a sub arbitrator OR, the bus arbitration module sends a request for common bus usage. The bus arbitrator grants control of the common bus to the IOC with highest priority by sending an IOC grant signal. This signal triggers the IOC grant latch, and in turn, the IOC common bus control module responds to the grant with a common bus OR signal from the originator ready latch and driver. About the same time, address information is placed onto the common bus. Once the address information is available, the IOC will wait for the ATR signal from the addressed device (in most cases, the memory interface module). This signal is applied to the grant latch to reset it. Once the address has been accepted by the addressed device, the memory busy signal becomes active and is sent to the common bus busy one-shots. The next response from the addressed device is the DTR signal to the IOC common bus control module. The DTR signal is then applied to the DTR control. Also at this time, the data is available on the common bus.

6.4.4 Common bus control (slave). When the IOC becomes a slave to an operation, the master device (after performing a handshake and priority determination routine similar to that

Source: https://assist.dla.mil -- Downloaded: 20261O-20T18:26Z
Check the source to verify that this is the currentversion before use.

described in 6.4.3) places an OR signal on the common bus and a function code. The IOC common bus control module responds with an ATR signal from the ATR flip-flop and a common bus busy signal. The busy signal is cleared by the clear busy - set DTR signal applied to the D-input of the DTR flip-flop. In the case of a read, DTR implies that data is on the bus, is stable, and is ready to be latched in. For a write, DTR removes data from the common bus. The DTR signal is sent from the IOC common bus control module to the master device.

6.4.5 Function control circuits. The function control circuits decode the common bus function code. The function code defines the operation to be executed.

6.4.6 Common bus timeout control circuits. The common bus has two types of timeouts: address timeout and data timeout. Address timeout is caused by a grant signal occurring and not subsequent ATR received within a 20 $\pm$ 4 microsecond window. Data timeouts are caused by:

(a) Busy going active and remaining that way for approximately 13 $\pm$ 3 microseconds.

(b) Busy going active and a DTR not occurring within 1 clock cycle.

(c) An ATR has occurred and busy has been inactive for 150 ns and does not go active within 250 ns.

The address timeout signal is latched into the timeout flip-flop and sent to originator control circuits to inform the master that the cycle did not complete within a specified time frame. The data timeout signal is also applied to the timeout flip-flop and sent to data timeout control circuits to clear the DTR flip-flop.

6.4.7 Clock start/stop control circuits. The clock start/stop control circuits are a series of AND-NOR gates and flip-flops used to generate wait request signals and ready control signals for the IOC arithmetic logic unit clock.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

6.4.8 Description of I/O pins.

Signal name	Pin number	I/O	Description
IOC SEL 0	2	I	IOC SEL 0
UI 4	4	I	UI 4
UI 5	5	I	UI 5
UI 6	6	I	UI 6
MCO CTRL FMT	9	I	MICRO CONTROL FORMAT
UI 0	13	I	UI 0
IOC SEL 1	14	I	IOC SEL 1
NO PROTECT	15	I	NO PROTECT
BYTE MODE	20	I	BYTE MODE
ENBL R/M/W	23	I	ENABLE READ/MODIFY/WRITE
WR DATA HIT	24	I	WRITE DATA HIT
ENBL CB LT	28	I	ENABLE CB LOOP TEST
GRANT (S.A.)	29	I	GRANT (S.A.)
BTE/BT (R/W)	32	I	BYTE/BIT (READ/WR LOWER)
INT C TP PRI	33	I	INTERRUPT CLASS TOP PRIORITY
ADDRESS 3	39	I	ADDRESS 3
COMM ADDR 2	48	I	COMMON ADDRESS 2
MAR/MDR B 1	50	I	MAR/MDR BUS 1
IOC GRANT	56	I	IOC GRANT
DATA TIMEOUT	57	I	DATA TIMEOUT

Source: https://assist.dla.mil -- Downloaded: 20260410-20T18:26Z
Check the source to verify that this is the current version before use.

6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
ADDR TIMEOUT	58	I	ADDR TIMEOUT
MAE IN	63	I	MAE IN
COMMAND BUSY	66	I	COMMAND BUSY (FROM HP IOC)
LD MAR	67	I	LD MAR
CLS III HIT	68	I	CLASS III HIT
MAIN ENBL SQ	70	I	MAINT ENABLE SEQ
UI 7	71	I	UI 7
ST ADDR BT 0	72	I	SET ADDRESS BIT 0
WAIT ACK	73	I	WAIT ACK
MPC MODE	74	I	MPC MODE
MAR/MDR B 0	76	I	MAR/MDR BUS 0
C2 (FRM GEN)	77	I	C2 (FROM GEN)
T3	79	I	T3
DATA HIT	80	I	DATA HIT
SET S.A. REQ	84	I	SET S.A. REQ (CHAIN + T INST)
BOOT DETECT	85	I	BOOT DETECT
T2	86	I	T2
MPC M+E+D	87	I	MPC MODE + ESA + DMA
MACLR	89	I	MACLR
BUS CLK	97	I	BUS CLK

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
C1 (FRM GEN)	98	I	C1 (FROM GEN)
BIAS	3	O	BIAS
LD CL I STK	7	O	LOAD CLASS I STACK
LD CL II STK	8	O	LOAD CLASS II STACK
LD CTRL WORD	11	O	LOAD CONTROL WORD
CLR CHAIN RQ	12	O	CLEAR CHAIN REQ
CK IOC CM RG	16	O	CLOCK IOC COMMAND REG
ENBL CB 0-9	17	O	ENABLE CB ADDRESS 0-9
COMM PEND	18	O	COMMAND PENDING
SUB ARB REQ	19	O	SUB ARB REQ
ENB MAR (CM)	21	O	ENB MAR (CM)
ENB MR (S/D)	22	O	ENABLE MAR (S/D, ESA, .DMA)
ADRS 1 (IRS)	26	O	ADRS 1 (INTRPT, READ SELECT)
ADRS 0 (IRS)	27	O	ADRS 0 (INTRPT, READ SELECT)
ENBL INT DTA	30	O	ENABLE INTRPT DATA (T.S.)
IOC WR PG RG	34	O	IOC WRITE PAGE REG (MAE)
EXEC PROTECT	35	O	EXECUTE PROTECT
RD PRO (MAE)	36	O	READ PROTECT (MAE)
WR PRO (MAE)	37	O	WRITE PROTECT (MAE)
ENBL MDR	38	O	ENBL MDR

Source: https://assist.dla.mil -- Downloaded: 20260410-20T18:26Z
Check the source to verify that this is the current version before use.

6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB NDRO DTCT	49	O	CB NDRO DETECT
S.A. OR	55	O	S.A. OR
ENL CB 10-15	59	O	ENABLE CB ADRS 10-15
ENL MAE INFO	62	O	ENL MAE INFO
DTR (CK MDR)	64	O	DTR (CLK MDR)
COM BSY	65	O	COMMAND BUSY (CP OR LP IOC)
SET CB D CYC	69	O	SET CB DATA CYCLE (N.U.)
S/D CLOCK	78	O	S/D CLOCK
D T/O-ATR RD	82	O	DATA T/O-ATR RCVD
CLK AD (T/O)	83	O	CLK ADDRESS (T/O)
GRANTED	88	O	GRANTED
RDY	91	O	READY
WAIT REQ	92	O	WAIT REQ
IO CLK 0-3	93	O	IO CLK 0-3
IO CLK 4-7	94	O	IO CLK 4-7
IO CLK 8-11	95	O	IO CLK 8-11
IO CLK 12-15	96	O	IO CLK 12-15
IO CK (IOCB)	99	O	IO CK (IOCB)
IO CK (IOCA)	100	O	IO CK (IOCA)
CB FCTN 0	41	I/O	CB FCTN 0 (ADRS 23)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

6.4.8 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CB FCTN 3	42	I/O	CB FCTN 3 (ADRS 26)
CB FCTN 2	43	I/O	CB FCTN 2 (ADRS 25)
CB FCTN 1	44	I/O	CB FCTN 1 (ADRS 24)
CB WR ENBL	45	I/O	CB WRITE ENABLE
CB AD/WR D0	46	I/O	CB ADRS/WRITE DATA BIT 0
CB AD/WR D1	47	I/O	CB ADRS/WRITE DATA BIT 1
CB OR	52	I/O	CB OR
CB DTR	53	I/O	CB DTR
CB BUSY	54	I/O	CB BUSY
CB ATR	61	I/O	CB ATR

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N722)

Agent:
NWSC Crane

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to	B to	Min	Max		
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

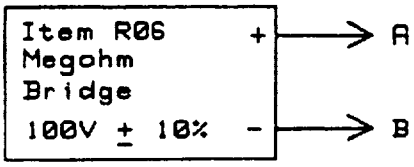
NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.



BWF

1,31,51,81	+5 V	IOC COMMON BUS CONTROL	
10,25,40,68,75,90	GND		
2	IOC SEL 0	XMIT	3
4	UI 4	LOAD CLASS I STACK	7
5	UI 5	LOAD CLASS II STACK	8
6	UI 6	LOAD CONTROL WORD	11
9	MICRO CONTROL FORMAT	CLR CHAIN REQ	12
13	UI 8	CLK IOC COMMAND REG	16
14	IOC SEL 1	ENBL CB ADDRESS 8-9	17
15	NO PROTECT	COMMAND PENDING	18
20	BYTE MODE	SUB ARB REQ	19
23	ENBL READ/MODIFY/WRITE	ENBL MAR (CM)	21
24	WRITE DATA HIT	ENBL MAR (S/D,ESA,DMA)	22
28	ENBL CB LOOP TEST	ADRS 1 (INTRPT READ SEL)	26
29	GRANT (S.A.)	ADRS 0 (INTRPT READ SEL)	27
32	BYTE BIT (R/W LOWER)	ENBL INTRPT DATA (T.S.)	30
33	INTRPT CLASS TOP PRIORITY	IOC WRITE PAGE REG (MAE)	34
39	ADDRESS 3	EXECUTE PROTECT	35
40	COMMON ADDRESS 2	READ PROTECT (MAE)	36
50	MAR/MDR BUS 1	WRITE PROTECT (MAE)	37
56	IOC GRANT	ENBL MDR	38
57	DATA TIMEOUT	CB NDRO DETECT	49
58	ADDRESS TIMEOUT	S.A. OR	55
63	MAE IN	ENBL CB ADDRESS 10-15	58
66	COMMAND BUSY (FROM HP IOC)	ENBL MAE INFO	62
67	LD MAR	DTR (CLK MDR)	64
69	CLASS III HIT	COMMAND BUSY (CP OR LP IOC)	65
70	MAINT ENABLE SEQ	SET CB DATA CYCLE (N.U.)	66
71	UI 7	S/D CLOCK	78
72	SET ADDRESS BIT 0	DATA T/O-ATR RCVD	82
73	WAIT ACK	CLK ADDRESS (T/O)	83
74	MPC MODE	GRANTED	89
76	MAR/MDR BUS 0	READY	91
77	C2 (FROM GEN)	WAIT REQ	92
78	T3	IO CLK 0-3	93
80	DATA HIT	IO CLK 4-7	94
84	SET S.A. REQ (CHAIN + T INST)	IO CLK 8-11	95
85	BOOT DETECT	IO CLK 12-15	96
86	T2	IO CLK (IOCB)	99
87	MPC MODE + ESA + DMA	IO CLK (IOCA)	100
89	MRCLR		
97	BUS CLK	CB FCTN 0 (ADRS 23)	41
98	C1 (FROM GEN)	CB FCTN 3 (ADRS 26)	42
		CB FCTN 2 (ADRS 25)	43
		CB FCTN 1 (ADRS 24)	44
		CB WRITE ENABLE	45
		CB ADRS WRITE/DATA BIT 0	46
		CB ADRS/WRITE DATA BIT 1	47
		CB OR	52
		CB DTR	53
		CB BUSY	54
		CB ATR	61

FIGURE 6. Isolation test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE III. Continuity test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, ground	10	25	0	1	Ohm	1-5
B2	Continuity, ground	10	40				
B3	Continuity, ground	60	75				
B4	Continuity, ground	60	90				
B5	Continuity, +5V	1	31				
B6	Continuity, +5V	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

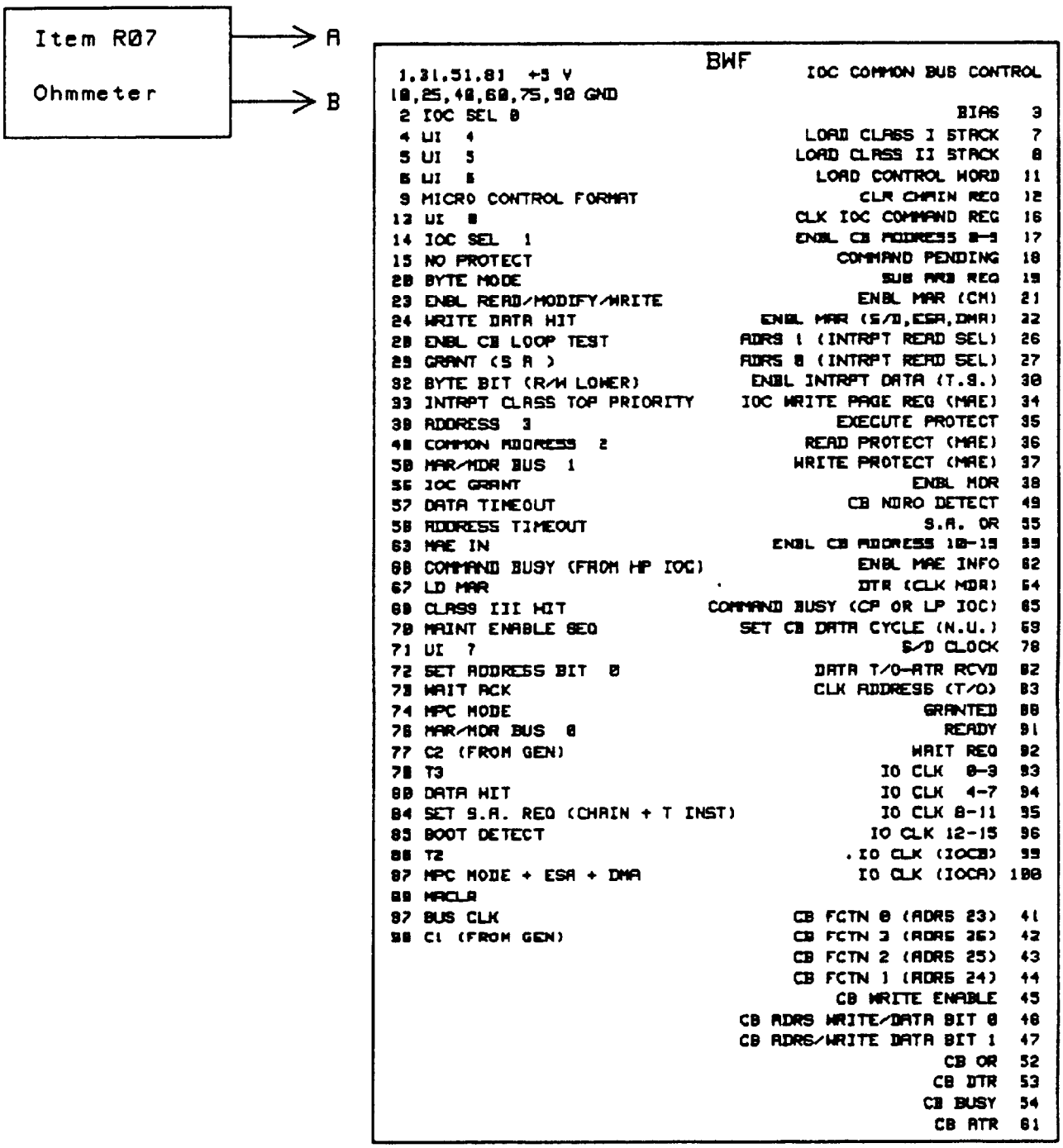


FIGURE 7. Continuity test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Notes
			Positive lead to	Negative lead to		
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.30 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:26Z
Check the source to verify that this is the current version before use.

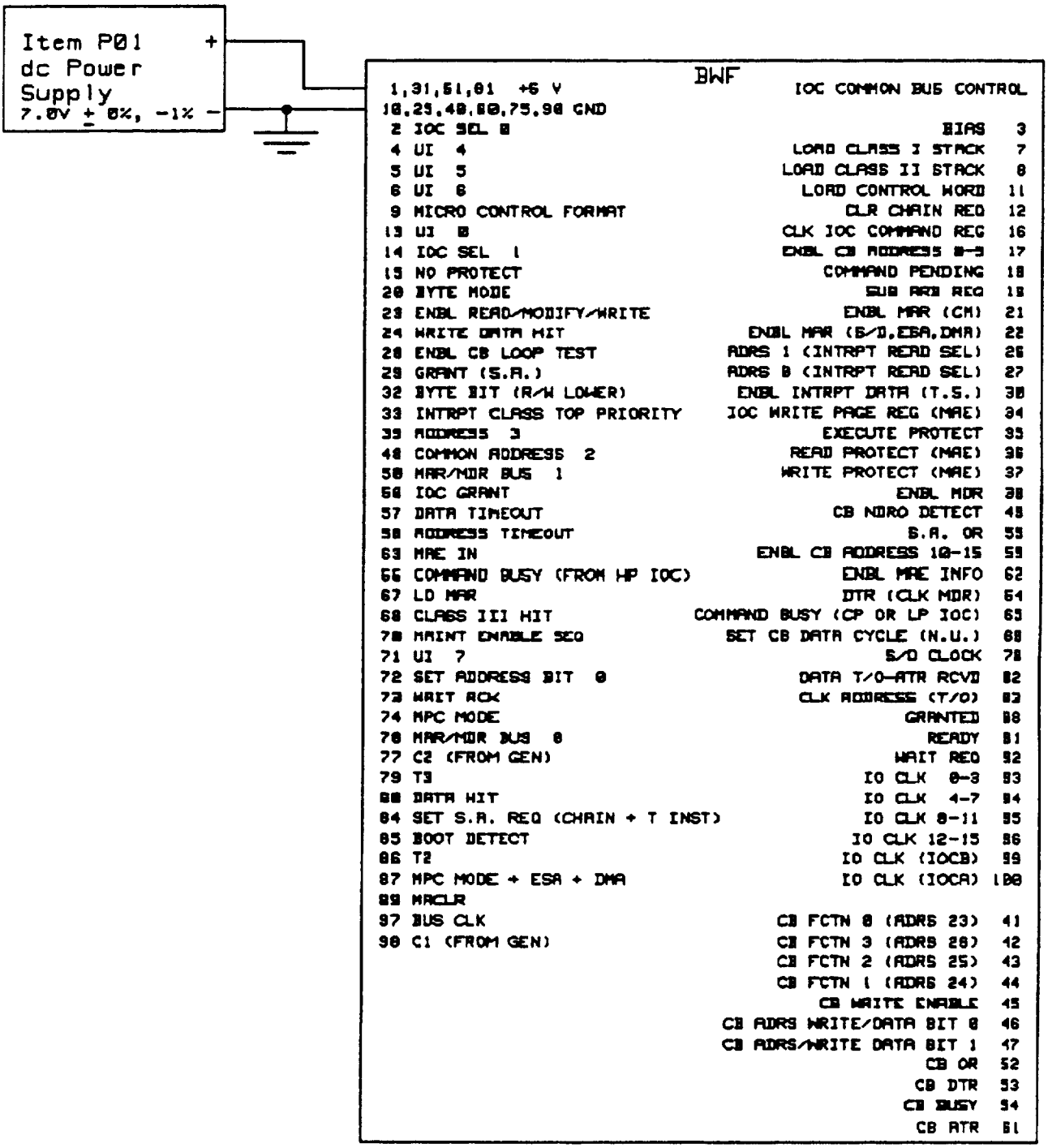


FIGURE 8. Vcc stress test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	IOC SEL 0	2	5.5	0.0	0.1	mA	1-3
D2	UI 4	4					
D3	UI 5	5					
D4	UI 6	6					
D5	MCO CTRL FMT	9					
D6	UI 0	13					
D7	IOC SEL 1	14					
D8	NO PROTECT	15	5.5	0.0	0.1	mA	1-3
D9	BYTE MODE	20	5.5	0.0	0.2	mA	1-3
D10	ENBL R/M/W	23	5.5	0.0	0.6	mA	1-3
D11	WR DATA HIT	24	5.5	0.0	0.1	mA	1-3
D12	ENBL CB LT	28	5.5	0.0	0.2	mA	1-3
D13	GRANT (S.A.)	29					
D14	BTE/BT (R/W)	32	5.5	0.0	0.2	mA	1-3
D15	INT C TP PRI	33	5.5	0.0	0.1	mA	1-3
D16	ADDRESS 3	39					
D17	COMM ADDR 2	48					
D18	MAR/MDR B 1	50	5.5	0.0	0.1	mA	1-3
D19	IOC GRANT	56	5.5	0.0	1.0	mA	1-3
D20	DATA TIMEOUT	57	5.5	0.0	0.1	mA	1-3
D21	ADDR TIMEOUT	58	5.5	0.0	0.1	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D22	MAE IN	63	5.5	-0.12	0.32	mA	1-3
D23	COMMAND BUSY	66	5.5	-0.12	0.22	mA	1-3
D24	LD MAR	67	5.5	0.0	0.1	mA	1-3
D25	CLS III HIT	68					
D26	MAIN ENBL SQ	70					
D27	UI 7	71	5.5	0.0	0.1	mA	1-3
D28	ST ADDR BT 0	72	5.5	0.0	1.0	mA	1-3
D29	WAIT ACK	73	5.5	0.0	0.1	mA	1-3
D30	MPC MODE	74	5.5	0.0	0.7	mA	1-3
D31	MAR/MDR B 0	76	5.5	0.0	2.0	mA	1-3
D32	C2 (FRM GEN)	77	5.5	0.0	1.0	mA	1-3
D33	T3	79	5.5	0.0	0.1	mA	1-3
D34	DATA HIT	80	5.5	0.0	1.0	mA	1-3
D35	SET S.A. REQ	84	5.5	0.0	0.2	mA	1-3
D36	BOOT DETECT	85	5.5	0.0	0.3	mA	1-3
D37	T2	86	5.5	0.0	0.1	mA	1-3
D38	MPC M+E+D	87	5.5	0.0	0.2	mA	1-3
D39	MACLR	89	5.5	0.0	0.1	mA	1-3
D40	BUS CLK	97	5.5	0.0	2.0	mA	1-3
D41	C1 (FRM GEN)	98	5.5	0.0	7.1	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D41
Power supply current limit	1.92 A	NA	D1-D41
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D41
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D41 D1-D41
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 5 uA ± 10 uA ± 30 uA ± 5 uA ± 10 uA ± 5 uA ± 50 uA ± 5 uA ± 22 uA ± 17 uA ± 5 uA ± 50 uA ± 5 uA ± 35 uA ± 100 uA ± 50 uA ± 5 uA ± 50 uA ± 10 uA ± 15 uA ± 5 uA ± 10 uA	D1-D8 D9 D10 D11 D12-D14 D15-D18 D19 D20,D21 D22 D23 D24-D27 D28 D29 D30 D31 D32 D33 D34 D35 D36 D37 D38

Source: https://assist.dla.mil -- Downloaded: 20260410-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA NA NA	± 5 uA ± 100 uA ± 360 uA	D39 D40 D41

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	BIAS	3	0.0	-150.0	-60.0	mA	1-4
E2	LD CL I STK	7					
E3	LD CL II STK	8					
E4	LD CTRL WORD	11					
E5	CLR CHAIN RQ	12	0.0	-150.0	-60.0	mA	1-4
E6	CK IOC CM RG	16	0.0	-100.0	-40.0	mA	1-4
E7	ENBL CB 0-9	17	0.0	-150.0	-60.0	mA	1-4
E8	COMM PEND	18					
E9	SUB ARB REQ	19					
E10	ENB MAR (CM)	21					
E11	ENB MR (S/D)	22	0.0	-150.0	-60.0	mA	1-4
E12	ADRS 1 (IRS)	26	0.0	-260.0	-30.0	mA	1-4
E13	ADRS 0 (IRS)	27	0.0	-260.0	-30.0	mA	1-4
E14	ENBL INT DTA	30	0.0	-150.0	-60.0	mA	1-4
E15	IOC WR PG RG	34	0.0	-325.0	-100.0	mA	1-4
E16	EXEC PROTECT	35					
E17	RD PRO (MAE)	36					
E18	WR PRO (MAE)	37					
E19	ENBL MDR	38					
E20	CB FCTN 0	41					
E21	CB FCTN 3	42	0.0	-325.0	-100.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	CB FCTN 2	43	0.0	-325.0	-100.0	mA	1-4
E23	CB FCTN 1	44					
E24	CB WR ENBL	45					
E25	CB AD/WR D0	46					
E26	CB AD/WR D1	47					
E27	CB NDRO DTCT	49					
E28	CB OR	52					
E29	CB DTR	53					
E30	CB BUSY	54					
E31	S.A. OR	55	0.0	-325.0	-100.0	mA	1-4
E32	ENL CB 10-15	59	0.0	-150.0	-60.0	mA	1-4
E33	CB ATR	61	0.0	-325.0	-100.0	mA	1-4
E34	ENL MAE INFO	62	0.0	-150.0	-60.0	mA	1-4
E35	DTR (CK MDR)	64	0.0	-100.0	-40.0	mA	1-4
E36	COM BSY	65	0.0	-150.0	-60.0	mA	1-4
E37	SET CB D CYC	69	0.0	-100.0	-40.0	mA	1-4
E38	S/D CLOCK	78	0.0	-100.0	-40.0	mA	1-4
E39	D T/O-ATR RD	82	0.0	-150.0	-60.0	mA	1-4
E40	CLK AD (T/O)	83	0.0	-230.0	-60.0	mA	1-4
E41	GRANTED	88	0.0	-150.0	-60.0	mA	1-4
E42	RDY	91	0.0	-150.0	-60.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E43	WAIT REQ	92	0.0	-150.0	-60.0	mA	1-4
E44	IO CLK 0-3	93	0.0	-225.0	-50.0	mA	1-4
E45	IO CLK 4-7	94					
E46	IO CLK 8-11	95					
E47	IO CLK 12-15	96					
E48	IO CK (IOCB)	99					
E49	IO CK (IOCA)	100	0.0	-225.0	-50.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E49
Power supply current limit	1.92 A	NA	E1-E49
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E49
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E49
low, (VL)	0.0 V	± 100 mV	E1-E49

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 4.50 mA	E1-E5
	NA	± 3.00 mA	E6
	NA	± 4.50 mA	E7-E11
	NA	± 11.5 mA	E12,E13
	NA	± 4.50 mA	E14
	NA	± 11.2 mA	E15-E31
	NA	± 4.50 mA	E32
	NA	± 11.2 mA	E33
	NA	± 4.50 mA	E34
	NA	± 3.00 mA	E35
	NA	± 4.50 mA	E36
	NA	± 3.00 mA	E37,E38
	NA	± 4.50 mA	E39
	NA	± 8.50 mA	E40
	NA	± 4.50 mA	E41-E43
	NA	± 8.75 mA	E44-E49

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

Source: https://assist.dla.mil -- Downloaded: 20260410-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F1	IOC SEL 0	2	2.4	-0.6	0.02	mA	1-3
F2	UI 4	4					
F3	UI 5	5					
F4	UI 6	6					
F5	MCO CTRL FMT	9					
F6	UI 0	13					
F7	IOC SEL 1	14					
F8	NO PROTECT	15	2.4	-0.6	0.02	mA	1-3
F9	BYTE MODE	20	2.4	-1.2	0.04	mA	1-3
F10	ENBL R/M/W	23	2.4	-3.6	0.12	mA	1-3
F11	WR DATA HIT	24	2.4	-0.6	0.02	mA	1-3
F12	ENBL CB LT	28	2.4	-2.2	0.04	mA	1-3
F13	GRANT (S.A.)	29	2.4	-1.2	0.04	mA	1-3
F14	BTE/BT (R/W)	32	2.4	-1.2	0.04	mA	1-3
F15	INT C TP PRI	33	2.4	-0.6	0.02	mA	1-3
F16	ADDRESS 3	39	2.4	-0.6	0.02	mA	1-3
F17	CB FCTN 0	41	2.4	-0.65	0.07	mA	1-3
F18	CB FCTN 3	42	2.4	-0.65	0.07	mA	1-3
F19	CB FCTN 2	43	2.4	-0.25	0.07	mA	1-3
F20	CB FCTN 1	44	2.4	-0.65	0.07	mA	1-3
F21	CB WR ENBL	45	2.4	-0.65	0.07	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 20260410-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
F22	CB AD/WR D0	46	2.4	-0.25	0.07	mA	1-3
F23	CB AD/WR D1	47	2.4	-0.25	0.07	mA	1-3
F24	COMM ADDR 2	48	2.4	-0.6	0.02	mA	1-3
F25	MAR/MDR B 1	50	2.4	-1.6	0.02	mA	1-3
F26	CB OR	52	2.4	-0.65	0.07	mA	1-3
F27	CB DTR	53					
F28	CB BUSY	54	2.4	-0.65	0.07	mA	1-3
F29	IOC GRANT	56	2.4	-2.0	0.05	mA	1-3
F30	DATA TIMEOUT	57	2.4	-0.6	0.02	mA	1-3
F31	ADDR TIMEOUT	58	2.4	-0.6	0.02	mA	1-3
F32	CB ATR	61	2.4	-0.65	0.07	mA	1-3
F33	MAE IN	63	2.4	-6.63	-2.83	mA	1-3
F34	COMMAND BUSY	66	2.4	-6.03	-2.85	mA	1-3
F35	LD MAR	67	2.4	-0.6	0.02	mA	1-3
F36	CLS III HIT	68					
F37	MAIN ENBL SQ	70					
F38	UI 7	71	2.4	-0.6	0.02	mA	1-3
F39	ST ADDR BT 0	72	2.4	-2.0	0.05	mA	1-3
F40	WAIT ACK	73	2.4	-0.6	0.02	mA	1-3
F41	MPC MODE	74	2.4	-4.4	0.18	mA	1-3
F42	MAR/MDR B 0	76	2.4	-4.0	0.1	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F43	C2 (FRM GEN)	77	2.4	-2.0	0.05	mA	1-3
F44	T3	79	2.4	-0.6	0.02	mA	1-3
F45	DATA HIT	80	2.4	-2.0	0.05	mA	1-3
F46	SET S.A. REQ	84	2.4	-0.8	0.04	mA	1-3
F47	BOOT DETECT	85	2.4	-3.8	0.06	mA	1-3
F48	T2	86	2.4	-0.6	0.02	mA	1-3
F49	MPC M+E+D	87	2.4	-1.2	0.04	mA	1-3
F50	MACLR	89	2.4	-0.6	0.02	mA	1-3
F51	BUS CLK	97	2.4	-0.8	0.1	mA	1-3
F52	C1 (FRM GEN)	98	2.4	-6.6	0.37	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F52
Power supply current limit	1.92 A	NA	F1-F52
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F52

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F52 F1-F52
Input current measurement	NA	± 31 uA	F1-F8
	NA	± 62 uA	F9
	NA	± 186 uA	F10
	NA	± 31 uA	F11
	NA	± 112 uA	F12
	NA	± 62 uA	F13,F14
	NA	± 31 uA	F15,F16
	NA	± 36 uA	F17,F18
	NA	± 16 uA	F19
	NA	± 36 uA	F20,F21
	NA	± 16 uA	F22,F23
	NA	± 31 uA	F24
	NA	± 81 uA	F25
	NA	± 36 uA	F26-F28
	NA	± 102 uA	F29
	NA	± 31 uA	F30,F31
	NA	± 36 uA	F32
	NA	± 190 uA	F33
	NA	± 159 uA	F34
	NA	± 31 uA	F35-F38
	NA	± 102 uA	F39
	NA	± 31 uA	F40
	NA	± 229 uA	F41
	NA	± 205 uA	F42
	NA	± 102 uA	F43
	NA	± 31 uA	F44
	NA	± 102 uA	F45
	NA	± 42 uA	F46
	NA	± 193 uA	F47
	NA	± 31 uA	F48
	NA	± 62 uA	F49
	NA	± 31 uA	F50
	NA	± 45 uA	F51
	NA	± 350 uA	F52

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the currentversion before use.

TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	IOC SEL 0	2	0.4	-0.6	0.02	mA	1-3
G2	UI 4	4					
G3	UI 5	5					
G4	UI 6	6					
G5	MCO CTRL FMT	9					
G6	UI 0	13					
G7	IOC SEL 1	14					
G8	NO PROTECT	15	0.4	-0.6	0.02	mA	1-3
G9	BYTE MODE	20	0.4	-1.2	0.04	mA	1-3
G10	ENBL R/M/W	23	0.4	-3.6	0.12	mA	1-3
G11	WR DATA HIT	24	0.4	-0.6	0.02	mA	1-3
G12	ENBL CB LT	28	0.4	-2.2	0.04	mA	1-3
G13	GRANT (S.A.)	29	0.4	-1.2	0.04	mA	1-3
G14	BTE/BT (R/W)	32	0.4	-1.2	0.04	mA	1-3
G15	INT C TP PRI	33	0.4	-0.6	0.02	mA	1-3
G16	ADDRESS 3	39	0.4	-0.6	0.02	mA	1-3
G17	CB FCTN 0	41	0.4	-0.65	0.07	mA	1-3
G18	CB FCTN 3	42	0.4	-0.65	0.07	mA	1-3
G19	CB FCTN 2	43	0.4	-0.25	0.07	mA	1-3
G20	CB FCTN 1	44	0.4	-0.65	0.07	mA	1-3
G21	CB WR ENBL	45	0.4	-0.65	0.07	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G22	CB AD/WR D0	46	0.4	-0.25	0.07	mA	1-3
G23	CB AD/WR D1	47	0.4	-0.25	0.07	mA	1-3
G24	COMM ADDR 2	48	0.4	-0.6	0.02	mA	1-3
G25	MAR/MDR B 1	50	0.4	-1.6	0.02	mA	1-3
G26	CB OR	52	0.4	-0.65	0.07	mA	1-3
G27	CB DTR	53					
G28	CB BUSY	54	0.4	-0.65	0.07	mA	1-3
G29	IOC GRANT	56	0.4	-2.0	0.05	mA	1-3
G30	DATA TIMEOUT	57	0.4	-0.6	0.02	mA	1-3
G31	ADDR TIMEOUT	58	0.4	-0.6	0.02	mA	1-3
G32	CB ATR	61	0.4	-0.65	0.07	mA	1-3
G33	MAE IN	63	0.4	-6.63	-2.83	mA	1-3
G34	COMMAND BUSY	66	0.4	-6.03	-2.85	mA	1-3
G35	LD MAR	67	0.4	-0.6	0.02	mA	1-3
G36	CLS III HIT	68					
G37	MAIN ENBL SQ	70					
G38	UI 7	71	0.4	-0.6	0.02	mA	1-3
G39	ST ADDR BT 0	72	0.4	-2.0	0.05	mA	1-3
G40	WAIT ACK	73	0.4	-0.6	0.02	mA	1-3
G41	MPC MODE	74	0.4	-4.4	0.18	mA	1-3
G42	MAR/MDR B 0	76	0.4	-4.0	0.1	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G43	C2 (FRM GEN)	77	0.4	-2.0	0.05	mA	1-3
G44	T3	79	0.4	-0.6	0.02	mA	1-3
G45	DATA HIT	80	0.4	-2.0	0.05	mA	1-3
G46	SET S.A. REQ	84	0.4	-0.8	0.04	mA	1-3
G47	BOOT DETECT	85	0.4	-3.8	0.06	mA	1-3
G48	T2	86	0.4	-0.6	0.02	mA	1-3
G49	MPC M+E+D	87	0.4	-1.2	0.04	mA	1-3
G50	MACLR	89	0.4	-0.6	0.02	mA	1-3
G51	BUS CLK	97	0.4	-0.8	0.1	mA	1-3
G52	C1 (FRM GEN)	98	0.4	-6.6	0.37	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G52
Power supply current limit	1.92 A	NA	G1-G52
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G52

Source: https://assist.dla.mil -- Downloaded: 20260410-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G52
low, (VL)	0.0 V	± 100 mV	G1-G52
Input current measurement	NA	± 31 uA	G1-G8
	NA	± 62 uA	G9
	NA	± 186 uA	G10
	NA	± 31 uA	G11
	NA	± 112 uA	G12
	NA	± 62 uA	G13,G14
	NA	± 31 uA	G15,G16
	NA	± 36 uA	G17,G18
	NA	± 16 uA	G19
	NA	± 36 uA	G20,G21
	NA	± 16 uA	G22,G23
	NA	± 31 uA	G24
	NA	± 81 uA	G25
	NA	± 36 uA	G26-G28
	NA	± 102 uA	G29
	NA	± 31 uA	G30,G31
	NA	± 36 uA	G32
	NA	± 190 uA	G33
	NA	± 159 uA	G34
	NA	± 31 uA	G35-G38
	NA	± 102 uA	G39
	NA	± 31 uA	G40
	NA	± 229 uA	G41
	NA	± 205 uA	G42
	NA	± 102 uA	G43
	NA	± 31 uA	G44
	NA	± 102 uA	G45
	NA	± 42 uA	G46
	NA	± 193 uA	G47
	NA	± 31 uA	G48
	NA	± 62 uA	G49
	NA	± 31 uA	G50
	NA	± 45 uA	G51
	NA	± 350 uA	G52

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	BIAS	3	-0.4	2.4	4.5	V	1-4
H2	LD CL I STK	7					
H3	LD CL II STK	8					
H4	LD CTRL WORD	11					
H5	CLR CHAIN RQ	12					
H6	CK IOC CM RG	16					
H7	ENBL CB 0-9	17					
H8	COMM PEND	18					
H9	SUB ARB REQ	19					
H10	ENB MAR (CM)	21					
H11	ENB MR (S/D)	22	-0.4	2.4	4.5	V	1-4
H12	ADRS 1 (IRS)	26	-0.16	2.4	4.5	V	1-4
H13	ADRS 0 (IRS)	27	-0.16	2.4	4.5	V	1-4
H14	ENBL INT DTA	30	-0.4	2.4	4.5	V	1-4
H15	IOC WR PG RG	34					
H16	EXEC PROTECT	35					
H17	RD PRO (MAE)	36					
H18	WR PRO (MAE)	37					
H19	ENBL MDR	38					
H20	CB FCTN 0	41					
H21	CB FCTN 3	42	-0.4	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H22	CB FCTN 2	43	-0.4	2.4	4.5	V	1-4
H23	CB FCTN 1	44					
H24	CB WR ENBL	45					
H25	CB AD/WR D0	46					
H26	CB AD/WR D1	47					
H27	CB NDRO DTCT	49					
H28	CB OR	52					
H29	CB DTR	53					
H30	CB BUSY	54					
H31	S.A. OR	55					
H32	ENL CB 10-15	59					
H33	CB ATR	61					
H34	ENL MAE INFO	62					
H35	DTR (CK MDR)	64					
H36	COM BSY	65					
H37	SET CB D CYC	69					
H38	S/D CLOCK	78					
H39	D T/O-ATR RD	82	-0.4	2.4	4.5	V	1-4
H40	CLK AD (T/O)	83	-0.16	2.4	4.5	V	1-4
H41	GRANTED	88	-0.4	2.4	4.5	V	1-4
H42	RDY	91	-0.4	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H43	WAIT REQ	92	-0.4	2.4	4.5	V	1-4
H44	IO CLK 0-3	93					
H45	IO CLK 4-7	94					
H46	IO CLK 8-11	95					
H47	IO CLK 12-15	96					
H48	IO CK (IOCB)	99					
H49	IO CK (IOCA)	100	-0.4	2.4	4.5	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H49
Power supply current limit	1.92 A	NA	H1-H49
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H11
	-0.16 mA	± 16 uA	H12,H13
	-0.4 mA	± 40 uA	H14-H39
	-0.16 mA	± 16 uA	H40
	-0.4 mA	± 40 uA	H41-H49

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H49 H1-H49
Output voltage measurement	NA	± 105 mV	H1-H49

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

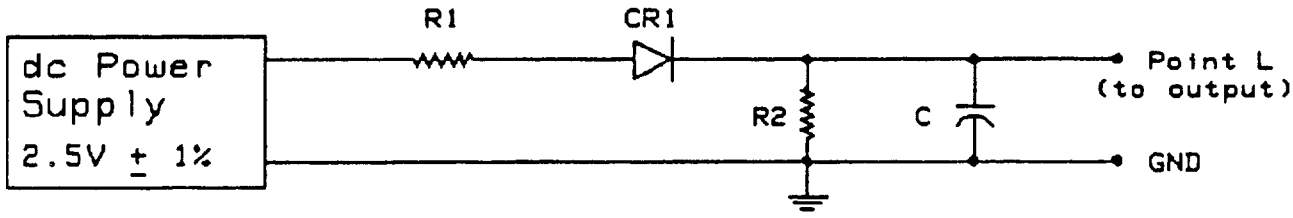
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.



NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 20261010-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
3	6200	NA	NA
7,8,11,12,16,17,18, 19,21,22,30,34,35, 36,37,38,41,42,43, 44,45,46,47,49,52, 53,54,55,59,61,62, 64,65,69,78,82,88, 91,92,93,94,95,96, 99,100	6200	6.0	0.5
26,27,83	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	LD CL I STK	7	6.0	0.0	0.5	V	1-4
J2	LD CL II STK	8					
J3	LD CTRL WORD	11					
J4	CLR CHAIN RQ	12					
J5	CK IOC CM RG	16					
J6	ENBL CB 0-9	17					
J7	COMM PEND	18					
J8	SUB ARB REQ	19					
J9	ENB MAR (CM)	21					
J10	ENB MR (S/D)	22	6.0	0.0	0.5	V	1-4
J11	ADRS 1 (IRS)	26	1.6	0.0	0.5	V	1-4
J12	ADRS 0 (IRS)	27	1.6	0.0	0.5	V	1-4
J13	ENBL INT DTA	30	6.0	0.0	0.5	V	1-4
J14	IOC WR PG RG	34					
J15	EXEC PROTECT	35					
J16	RD PRO (MAE)	36					
J17	WR PRO (MAE)	37					
J18	ENBL MDR	38					
J19	CB FCTN 0	41					
J20	CB FCTN 3	42					
J21	CB FCTN 2	43	6.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J22	CB FCTN 1	44	6.0	0.0	0.5	V	1-4
J23	CB WR ENBL	45					
J24	CB AD/WR D0	46					
J25	CB AD/WR D1	47					
J26	CB NDRO DTCT	49					
J27	CB OR	52					
J28	CB DTR	53					
J29	CB BUSY	54					
J30	S.A. OR	55					
J31	ENL CB 10-15	59					
J32	CB ATR	61					
J33	ENL MAE INFO	62					
J34	DTR (CK MDR)	64					
J35	COM BSY	65					
J36	SET CB D CYC	69					
J37	S/D CLOCK	78					
J38	D T/O-ATR RD	82	6.0	0.0	0.5	V	1-4
J39	CLK AD (T/O)	83	1.6	0.0	0.5	V	1-4
J40	GRANTED	88	6.0	0.0	0.5	V	1-4
J41	RDY	91					
J42	WAIT REQ	92	6.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J43	IO CLK 0-3	93	6.0	0.0	0.5	V	1-4
J44	IO CLK 4-7	94					
J45	IO CLK 8-11	95					
J46	IO CLK 12-15	96					
J47	IO CK (IOCB)	99					
J48	IO CK (IOCA)	100	6.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J48
Power supply current limit	1.92 A	NA	J1-J48
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA 1.6 mA 6.0 mA	± 600 UA ± 160 uA ± 600 uA ± 160 uA ± 600 uA	J1-J10 J11,J12 J13-J38 J39 J40-J48
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J48 J1-J48

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage measurement	NA	± 25 mV	J1-J48

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
High level							
K1	IOC WR PG RG	34	2.4	-0.05	0.05	mA	1-3
K2	EXEC PROTECT	35					
K3	RD PRO (MAE)	36					
K4	WR PRO (MAE)	37					
K5	CB NDRO DTCT	49					
K6	S.A. OR	55	2.4	-0.05	0.05	mA	1-3
Low level							
K7	IOC WR PG RG	34	0.4	-0.05	0.05	mA	1-3
K8	EXEC PROTECT	35					
K9	RD PRO (MAE)	36					
K10	WR PRO (MAE)	37					
K11	CB NDRO DTCT	49					
K12	S.A. OR	55	0.4	-0.05	0.05	mA	1-3

NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Power supply current limit	1.92 A	NA	K1-K12
Forcing voltage, (Voh) (Vol)	2.4 V 0.4 V	± 50 mV ± 50 mV	K1-K6 K7-K12
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K1-K12 K1-K12
Output current measurement	NA	± 5 uA	K1-K12

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 20261-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	IOC SEL 0	2	-18.0	-1.2	0.0	V	1-3
L2	UI 4	4					
L3	UI 5	5					
L4	UI 6	6					
L5	MCO CTRL FMT	9					
L6	UI 0	13					
L7	IOC SEL 1	14					
L8	NO PROTECT	15					
L9	BYTE MODE	20					
L10	ENBL R/M/W	23					
L11	WR DATA HIT	24					
L12	ENBL CB LT	28					
L13	GRANT (S.A.)	29					
L14	BTE/BT (R/W)	32					
L15	INT C TP PRI	33					
L16	ADDRESS 3	39					
L17	COMM ADDR 2	48					
L18	MAR/MDR B 1	50					
L19	IOC GRANT	56					
L20	DATA TIMEOUT	57					
L21	ADDR TIMEOUT	58	-18.0	-1.2	0.0	V	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L22	LD MAR	67	-18.0	-1.2	0.0	V	1-3
L23	CLS III HIT	68					
L24	MAIN ENBL SQ	70					
L25	UI 7	71					
L26	ST ADDR BT 0	72					
L27	WAIT ACK	73					
L28	MPC MODE	74					
L29	MAR/MDR B 0	76					
L30	C2 (FRM GEN)	77					
L31	T3	79					
L32	DATA HIT	80					
L33	SET S.A. REQ	84					
L34	BOOT DETECT	85					
L35	T2	86					
L36	MPC M+E+D	87					
L37	MACLR	89					
L38	BUS CLK	97					
L39	C1 (FRM GEN)	98	-18.0	-1.2	0.0	V	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L39
Power supply current limit	NA	NA	L1-L39
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L39
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L39 L1-L39
Input voltage measurement	NA	± 60 mV	L1-L39

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	6.3	11.7	uF	1-4
M2	PSFC	51	5.6	10.4	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

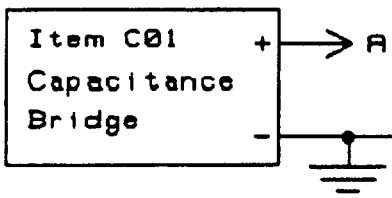
Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 270 nF ± 240 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.



BWF

1,31,51,81 +5 V		IOC COMMON BUS CONTROL	
10,25,40,60,75,90 GND			
2 IOC SEL 0		BIAS	3
4 UI 4		LOAD CLASS I STACK	7
5 UI 5		LOAD CLASS II STACK	8
6 UI 6		LOAD CONTROL WORD	11
8 MICRO CONTROL FORMAT		CLR CHAIN REQ	12
13 UI 0		CLK IOC COMMAND REG	16
14 IOC SEL 1		ENBL CB ADDRESS 8-9	17
15 NO PROTECT		COMMAND PENDING	18
20 BYTE MODE		SUB ARB REQ	19
23 ENBL READ/MODIFY/WRITE		ENBL MAR (CM)	21
24 WRITE DATA HIT		ENBL MAR (S/D,ESA,DMA)	22
28 ENBL CB LOOP TEST		ADRS 1 (INTRPT READ SEL)	26
29 GRANT (S.A.)		ADRS 0 (INTRPT READ SEL)	27
32 BYTE BIT (R/W LOWER)		ENBL INTRPT DATA (T.S.)	30
33 INTRPT CLASS TOP PRIORITY		IOC WRITE PAGE REG (MAE)	34
39 ADDRESS 3		EXECUTE PROTECT	35
40 COMMON ADDRESS 2		READ PROTECT (MAE)	36
50 MAR/MDR BUS 1		WRITE PROTECT (MAE)	37
56 IOC GRANT		ENBL MDR	38
57 DATA TIMEOUT		CB NDRO DETECT	49
58 ADDRESS TIMEOUT		S.A. OR	55
63 MAE IN		ENBL CB ADDRESS 10-15	58
66 COMMAND BUSY (FROM HP IOC)		ENBL MAE INFO	62
67 LD MAR		DTR (CLK MDR)	64
68 CLASS III HIT		COMMAND BUSY (CP OR LP IOC)	65
70 MAINT ENABLE SEQ		SET CB DATA CYCLE (N.U.)	69
71 UI 7		S/D CLOCK	70
72 SET ADDRESS BIT 0		DATA T/O-ATR RCVD	82
73 WAIT ACK		CLK ADDRESS (T/O)	83
74 MPC MODE		GRANTED	86
76 MAR/MDR BUS 0		READY	91
77 C2 (FROM GEN)		WAIT REQ	92
78 T3		IO CLK 0-3	93
80 DATA HIT		IO CLK 4-7	94
84 SET S.A. REQ (CHAIN + T INST)		IO CLK 8-11	95
85 BOOT DETECT		IO CLK 12-15	96
86 T2		IO CLK (IOCB)	99
87 MPC MODE + ESA + DMA		IO CLK (IOCA)	100
89 MACLR			
97 BUS CLK		CB FCTN 0 (ADRS 23)	41
98 C1 (FROM GEN)		CB FCTN 3 (ADRS 26)	42
		CB FCTN 2 (ADRS 25)	43
		CB FCTN 1 (ADRS 24)	44
		CB WRITE ENABLE	45
		CB ADRS WRITE/DATA BIT 0	46
		CB ADRS/WRITE DATA BIT 1	47
		CB OR	52
		CB DTR	53
		CB BUSY	54
		CB ATR	61

FIGURE 10. Power supply filter capacitance test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	535	1284	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.92 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 37.4 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.92 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

$$\text{High level: } 2.4\ \text{V} \leq \text{Voh} \leq \text{Vcc}$$
$$\text{Low level: } 0.0\ \text{V} \leq \text{Vol} \leq 0.5\ \text{V}$$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

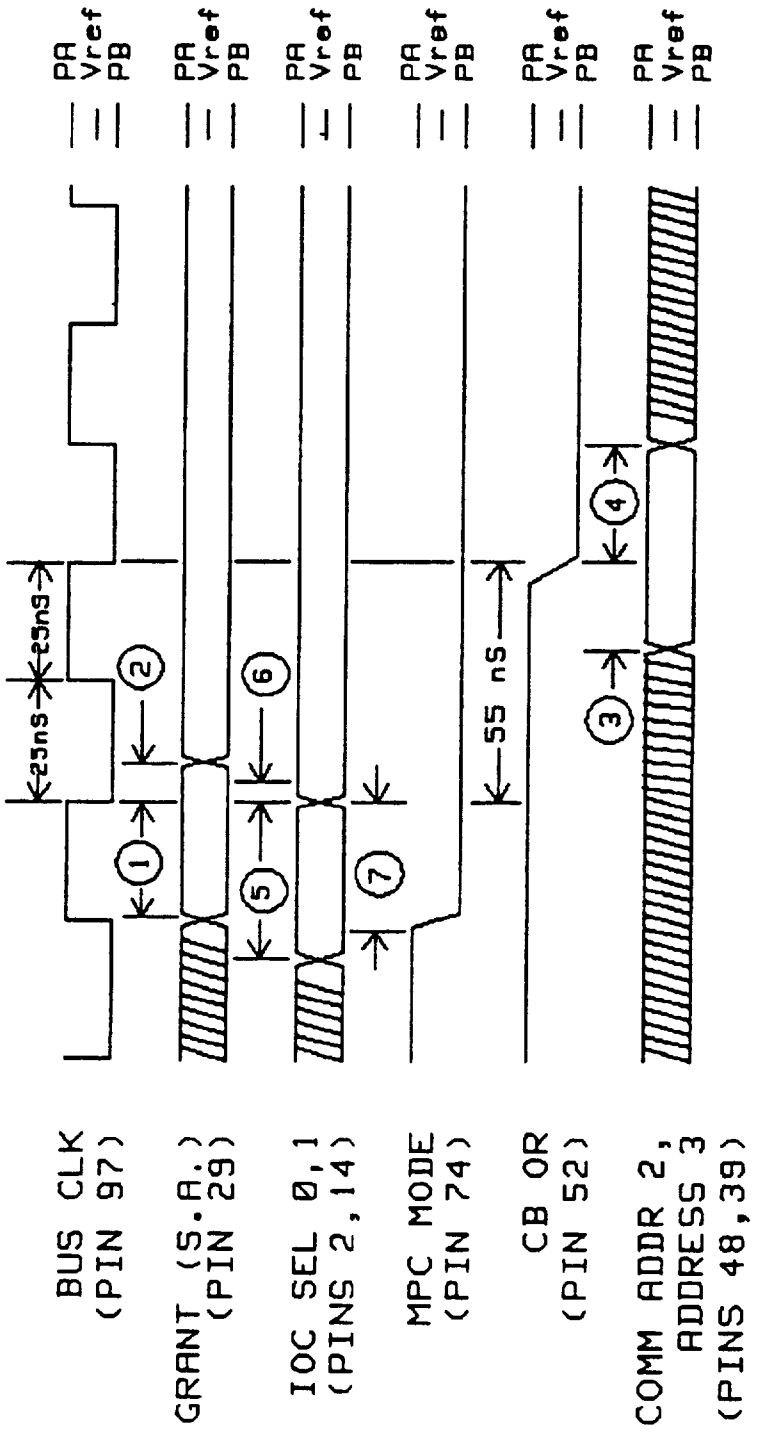


FIGURE 11a. Input timing parameters for table XV.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:26Z
Check the source to verify that this is the currentversion before use.

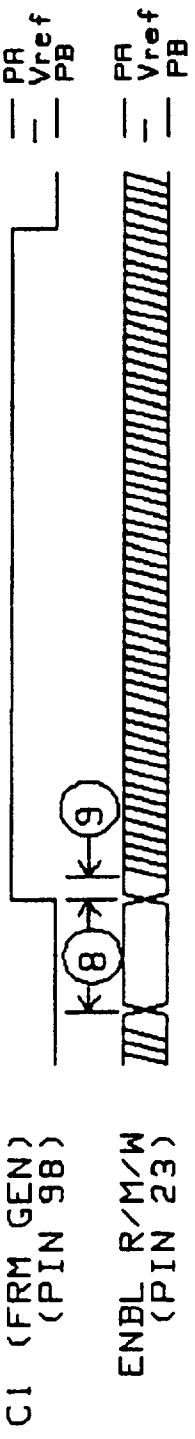


FIGURE 11b. Input timing parameters for table XV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES:

1. (1) GRANT (S.A.) to BUS CLK. Setup time: 23 nS minimum.

2. (2) BUS CLK to GRANT (S.A.). Hold time: 6 nS minimum.

3. (3) COMM ADDR 2, ADDRESS 3 to BUS CLK. Setup time: 20 nS minimum.

4. (4) BUS CLK to COMM ADDR 2, ADDRESS 3. Hold time: 26 nS minimum.

5. (5) IOC SEL 0, IOC SEL 1 to BUS CLK. Setup time: 35 nS minimum.

6. (6) BUS CLK to IOC SEL 0, IOC SEL 1. Hold time: 3 nS minimum.

7. (7) MPC MODE to BUS CLK. Setup time: 23 nS minimum.

8. (8) ENBL R/M/W to C1 (FRM GEN). Setup time: 39 nS minimum.

9. (9) C1 (FRM GEN) to ENBL R/M/W. Hold time: 3 nS minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.92 A	NA	Q1-Q4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage			
high, (VH)	2.4 V	± 100 mV	Q1-Q4
low, (VL)	0.4 V	± 100 mV	Q1-Q4
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

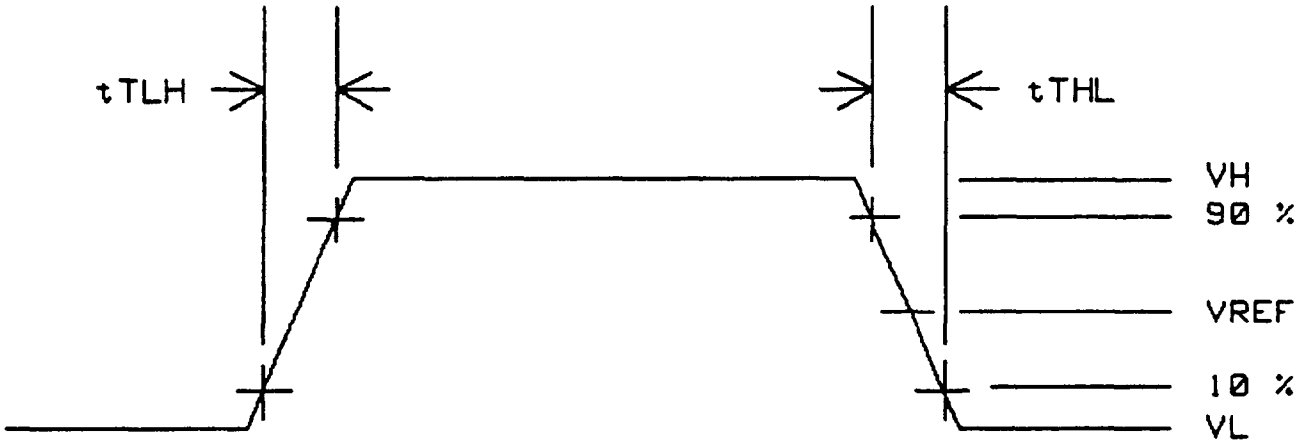
(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL ≤ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tsu	GRANT (S.A.) to BUS CLK	23	23	23	--	--	--	nS	12
2	th	BUS CLK to GRANT (S.A.)	6	6	6	--	--	--	nS	12
3	tsu	COMM ADDR 2 to BUS CLK	20	20	20	--	--	--	nS	12
3	tsu	ADDRESS 3 to BUS CLK	20	20	20	--	--	--	ns	12
4	th	BUS CLK to COMM ADDR 2	26	26	26	--	--	--	nS	12
4	th	BUS CLK to ADDRESS 3	26	26	26	--	--	--	nS	12
5	tsu	IOC SEL 0 to BUS CLK	35	35	35	--	--	--	nS	12
5	tsu	IOC SEL 1 to BUS CLK	35	35	35	--	--	--	nS	12
6	th	BUS CLK to IOC SEL 0	3	3	3	--	--	--	nS	12
6	th	BUS CLK to IOC SEL 1	3	3	3	--	--	--	nS	12
7	tsu	MPC MODE to BUS CLK	23	23	23	--	--	--	nS	12
8	tsu	ENBL R/M/W to C1 (FRM GEN)	39	39	39	--	--	--	nS	12
9	th	C1 (FRM GEN) to ENBL R/M/W	3	3	3	--	--	--	nS	12
10	tpd	BUS CLK to SUB ARB REQ	--	--	--	27	27	27	nS	12

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters. - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	BUS CLK to CK IOC CM RG	--	--	--	39	39	39	nS	12
11	tpd	BUS CLK to COM BSY	--	--	--	39	39	39	nS	12
12	tpd	BUS CLK to ENB MAR (CM)	--	--	--	41	41	41	nS	12
12	tpd	BUS CLK to ENB MR (S/D)	--	--	--	41	41	41	nS	12
12	tpd	BUS CLK to ENBL MDR	--	--	--	41	41	41	nS	12
13	tpd	BUS CLK to CB ATR	--	--	--	34	34	34	nS	12
14	tpd	BUS CLK to ENBL CB 0-9	--	--	--	39	39	39	nS	12
15	tpd	BUS CLK to ADRS 0 (IRS)	--	--	--	61	61	61	nS	12
15	tpd	BUS CLK to ADRS 1 (IRS)	--	--	--	61	61	61	nS	12
16	tpd	GRANT (S.A.) to ENL MAE INFO	--	--	--	25	25	25	nS	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

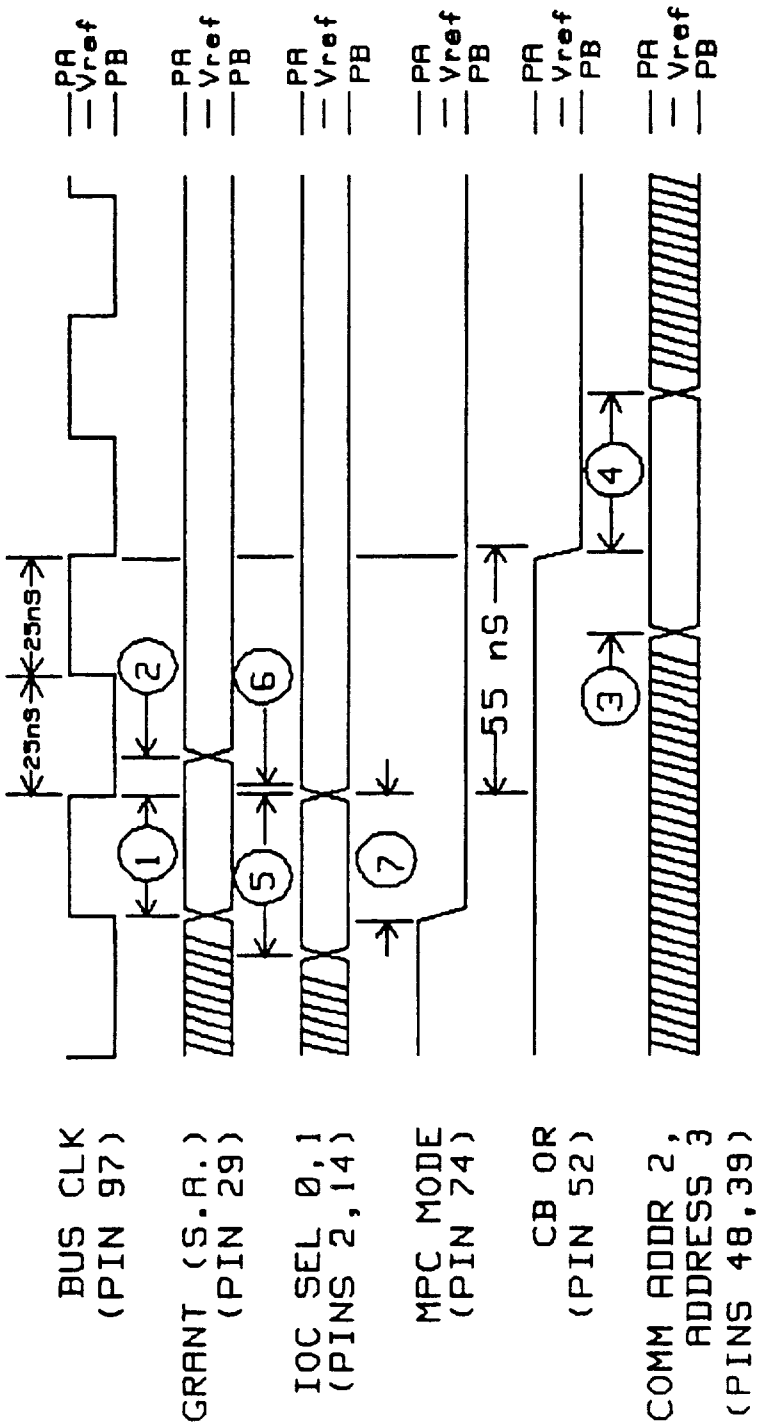


FIGURE 12a. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

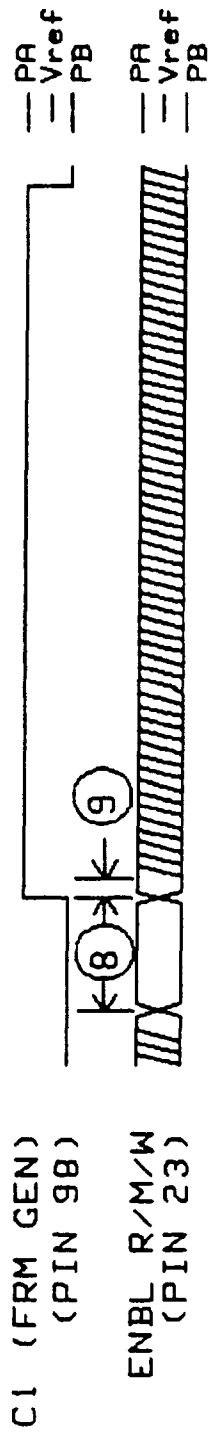


FIGURE 12b. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

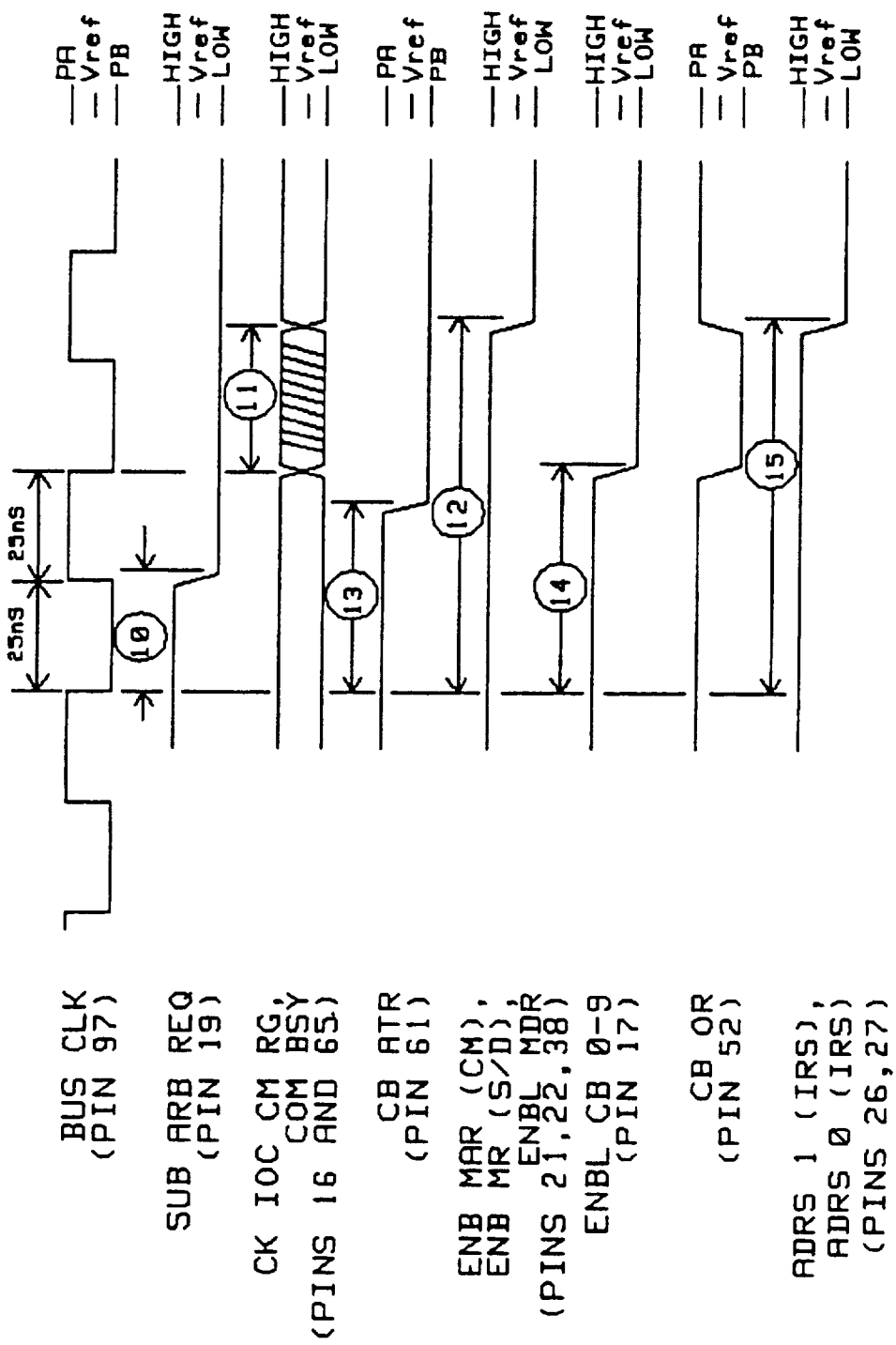


FIGURE 12c. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

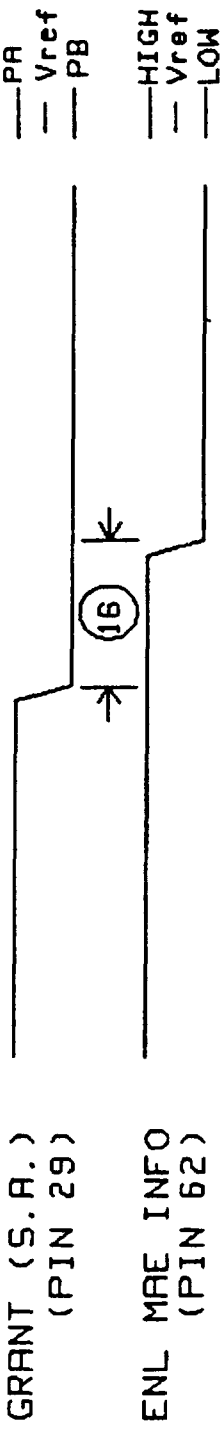


FIGURE 12d. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	IOC SEL 0	2	2	75	pF	1-4
R2	UI 4	4	2	50	pF	1-4
R3	UI 5	5				
R4	UI 6	6	2	50	pF	1-4
R5	MCO CTRL FMT	9	2	75	pF	1-4
R6	UI 0	13	2	50	pF	1-4
R7	IOC SEL 1	14	2	75	pF	1-4
R8	NO PROTECT	15	2	75	pF	1-4
R9	BYTE MODE	20	4	75	pF	1-4
R10	ENBL R/M/W	23	10	175	pF	1-4
R11	WR DATA HIT	24	2	50	pF	1-4
R12	ENBL CB LT	28	4	125	pF	1-4
R13	GRANT (S.A.)	29	4	75	pF	1-4
R14	BTE/BT (R/W)	32	4	100	pF	1-4
R15	INT C TP PRI	33	2	75	pF	1-4
R16	ADDRESS 3	39				
R17	COMM ADDR 2	48	2	75	pF	1-4
R18	MAR/MDR B 1	50	2	50	pF	1-4
R19	IOC GRANT	56				
R20	DATA TIMEOUT	57	2	50	pF	1-4

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	ADDR TIMEOUT	58	2	50	pF	1-4
R22	MAE IN	63	6	100	pF	1-4
R23	COMMAND BUSY	66	2	100	pF	1-4
R24	LD MAR	67	2	50	pF	1-4
R25	CLS III HIT	68	2	50	pF	1-4
R26	MAIN ENBL SQ	70	2	75	pF	1-4
R27	UI 7	71	2	100	pF	1-4
R28	ST ADDR BT 0	72	2	50	pF	1-4
R29	WAIT ACK	73	2	75	pF	1-4
R30	MPC MODE	74	14	175	pF	1-4
R31	MAR/MDR B 0	76	4	50	pF	1-4
R32	C2 (FRM GEN)	77	2	50	pF	1-4
R33	T3	79				
R34	DATA HIT	80	2	50	pF	1-4
R35	SET S.A. REQ	84	2	100	pF	1-4
R36	BOOT DETECT	85	6	125	pF	1-4
R37	T2	86	2	75	pF	1-4
R38	MPC M+E+D	87	4	125	pF	1-4
R39	MACLR	89	2	50	pF	1-4
R40	BUS CLK	97	4	75	pF	1-4
R41	C1 (FRM GEN)	98	16	150	pF	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Bidirectionals						
R42	CB FCTN 0	41	4	75	pF	1-4
R43	CB FCTN 3	42	4	100	pF	1-4
R44	CB FCTN 2	43	4	75	pF	1-4
R45	CB FCTN 1	44				
R46	CB WR ENBL	45				
R47	CB AD/WR D0	46				
R48	CB AD/WR D1	47	4	75	pF	1-4
R49	CB OR	52	2	100	pF	1-4
R50	CB DTR	53	2	75	pF	1-4
R51	CB BUSY	54				
R52	CB ATR	61	2	75	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R52
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R52

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 3.65 pF	R1
	NA	± 2.4 pF	R2-R4
	NA	± 3.65 pF	R5
	NA	± 2.4 pF	R6
	NA	± 3.65 pF	R7,R8
	NA	± 3.55 pF	R9
	NA	± 8.25 pF	R10
	NA	± 2.4 pF	R11
	NA	± 6.05 pF	R12
	NA	± 3.55 pF	R13
	NA	± 4.8 pF	R14
	NA	± 3.65 pF	R15-R17
	NA	± 2.4 pF	R18-R21
	NA	± 4.7 pF	R22
	NA	± 4.9 pF	R23
	NA	± 2.4 pF	R24,R25
	NA	± 3.65 pF	R26
	NA	± 4.9 pF	R27
	NA	± 2.4 pF	R28
	NA	± 3.65 pF	R29
	NA	± 8.05 pF	R30
	NA	± 2.3 pF	R31
	NA	± 2.4 pF	R32-R34
	NA	± 4.9 pF	R35
	NA	± 5.95 pF	R36
	NA	± 3.65 pF	R37
	NA	± 6.05 pF	R38
	NA	± 2.4 pF	R39
	NA	± 3.55 pF	R40
	NA	± 6.7 pF	R41
	NA	± 3.55 pF	R42
	NA	± 4.8 pF	R43
	NA	± 3.55 pF	R44-R48
	NA	± 4.9 pF	R49
	NA	± 3.65 pF	R50-R52

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

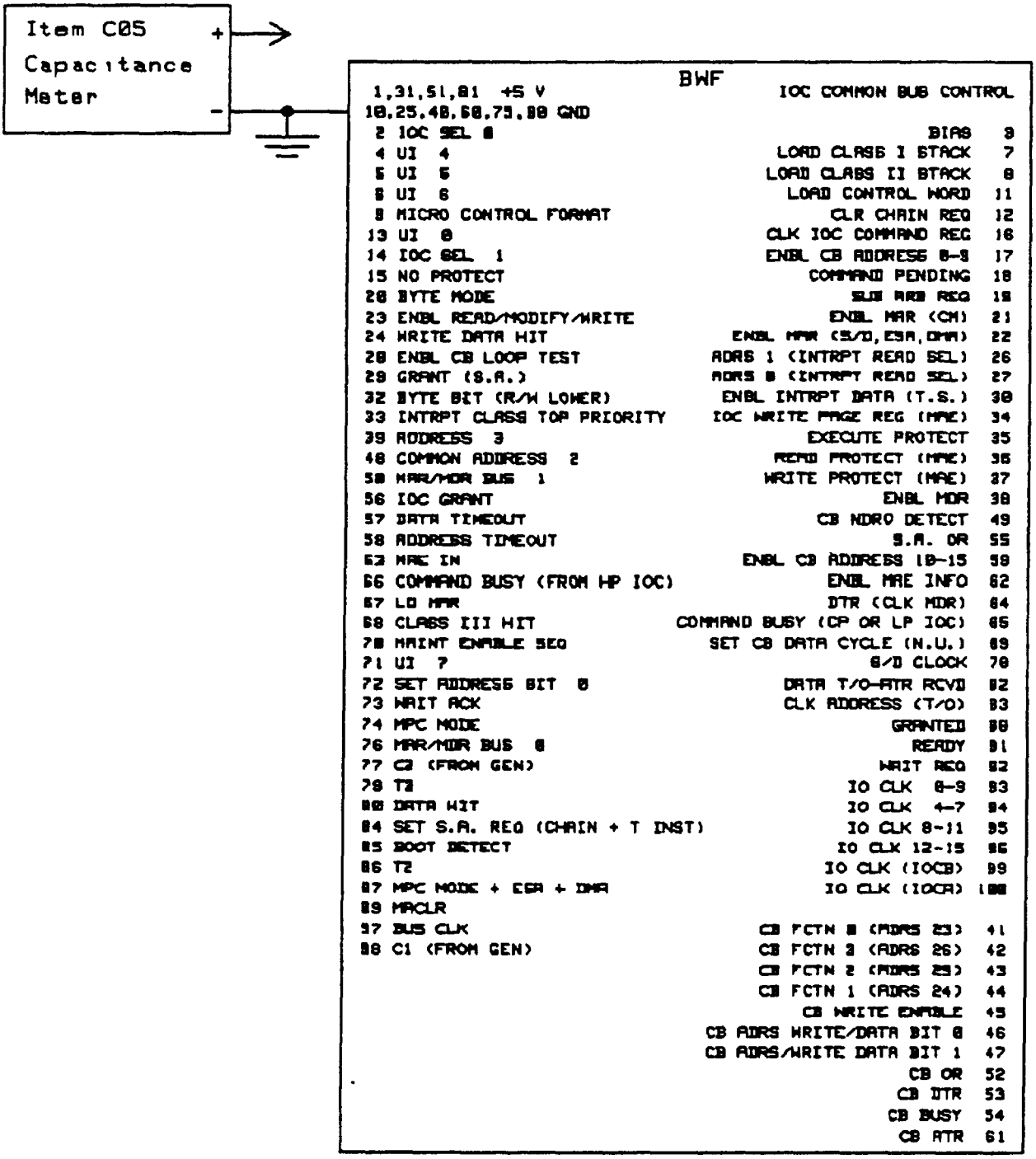


FIGURE 13. Input and output capacitance test circuit for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XIX. Input clamp diode latch up test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
S1	LD MAR	67	-18.0	-1.2	2.4	V	1-3
S2	C1 (FRM GEN)	98	-18.0	-1.2	2.4	V	1-3

NOTES:

1. Input clamp diode latch up test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	S1,S2
Power supply current limit	1.92 A	NA	S1,S2
Forcing current, (Iin)	-18 mA	± 1.8 mA	S1,S2
Pattern voltage high, (VH) low, (VL)	NA NA	NA NA	S1,S2 S1,S2
Input voltage measurement	NA	± 180 mV	S1,S2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XIX and the voltage then measured.

(d) The forcing pulse shall have a duration of 500 ± 20 nS.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.

TABLE XX. Power supply transient amplitude test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1	0	400	mV	1-4	14,15
T2	PSTA	51	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1.92 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T18:26Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

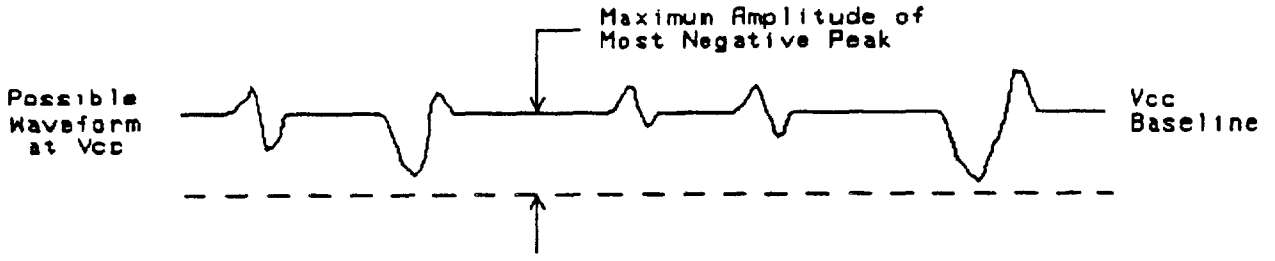
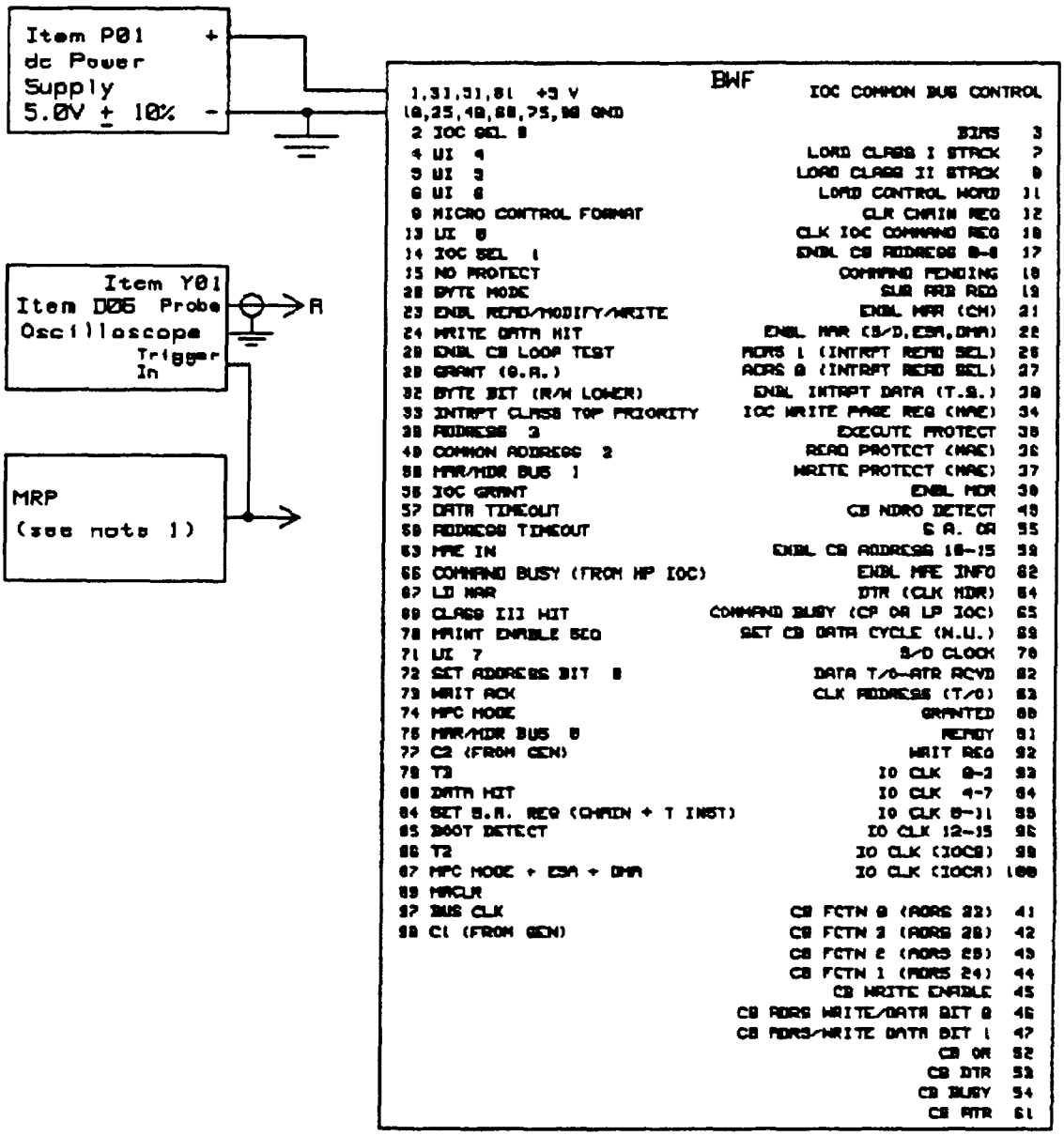


FIGURE 14. Power supply transient amplitude waveform for table XX.

Source: https://assist.dla.mil -- Downloaded: 202640-20T18:26Z
Check the source to verify that this is the current version before use.



NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-M-28787/511	Modules, Standard Electronic, IOC Common Bus Control Module, Key Code BWF

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 20261020T18:26Z
Check the source to verify that this is the current version before use.

(Fold along this line)

DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 5523)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 5523)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:26Z
Check the source to verify that this is the current version before use.